

Quadrotor Gray-box Model Identification from High-Speed Flight Data

Sihao Sun^{*}, Coen C. de Visser[†] and Qiping Chu[‡]
Delft University of Technology, 2629 HS Delft, The Netherlands

In order to explore the aerodynamic effects on a quadrotor in the high-speed flight regime and establish an accurate nonlinear model, free flight tests with a quadrotor have been carried out in a large-scale wind tunnel. The flight data reveal that complex aerodynamic interactions could appear and significantly influence the forces and moments acting on the quadrotor, which indicates the inaccuracy of state-of-art models established based on helicopter aerodynamic theory. To cope with this problem, gray-box models considering these effects are identified from flight data using a stepwise system identification approach, which combines both prior-knowledge of rotorcraft aerodynamic properties as well as data observations. Previous models introduced in the literature are compared with the gray-box models. Validation results show an 80% reduction of moment model residuals and 20% reduction of force model residuals.

Nomenclatures

α, β	=	angle of attack and sideslip angle, rad
ρ	=	air density, kg/m ³
g	=	acceleration of gravity, m/s ²
V, V_g, V_{wind}	=	airspeed, ground speed and wind speed, m/s.
u, v, w	=	airspeed components in the body frame, m/s
Ω, p, q, r	=	body angular rates, rad/s
$I_v, I_x, I_y, I_z, I_{xz}$	=	inertia moment of the vehicle, kg·m ²
I_p	=	inertia moment of the propeller, kg·m ²
m	=	total mass of the vehicle, kg
b, l, R	=	vehicle geometry parameters, rotor radius, m
κ_0, τ_0	=	force and torque coefficient of the hovering model, N·s, N·m·s
λ_r	=	damping rate in the hovering model, N·m·s
R_{BG}	=	rotational matrix from the ground frame to the body frame

^{*}Ph.D Student, Control and Simulation Section, Faculty of Aerospace Engineering, Kluyverweg 1.

[†]Assistant Professor, Control and Simulation Section, Faculty of Aerospace Engineering, Kluyverweg 1. Member AIAA.

[‡]Associate Professor, Control and Simulation Section, Faculty of Aerospace Engineering, Kluyverweg 1.

F, F_x, F_y, F_z	= aerodynamic forces, N
M, M_x, M_y, M_z	= aerodynamic moments, N·m
M_h, F_h	= aerodynamic force and moment estimation from the hovering model, N and N·m respectively
M_{I_p}	= rotor inertia related moment
T	= thrust, N
C_t	= thrust coefficient
$C_x, C_y, C_z, C_l, C_m, C_n$	= aerodynamic forces and moments coefficients
C_h	= horizontal force coefficient
Ω_i	= rotation speed of the i th rotor, rad/s
$\bar{\Omega}$	= geometric average of rotor speeds, rad/s
ω_i	= normalized rotation speed of the i th rotor
u_p, u_q, u_r	= control inputs for roll, pitch and yaw
N	= number of rotors, or, number of measurement samples
μ_x, μ_y, μ_z	= advance ratios
μ_h	= horizontal advanced ratio
$\bar{p}, \bar{q}, \bar{r}$	= normalized body angular rates
z	= force and moment measurements
y	= model outputs
ϵ	= model residual
A	= regressor matrix
$\theta, \hat{\theta}$	= vector of model parameters and its estimation
$S(\cdot)$	= candidate set
x	= vector of model independent variables
$P^d(x)$	= set comprising all basis of the d th order polynomial with x as the independent variable
n	= dimension of x
$\hat{d}$	= number of terms in $P^d(x)$
ξ	= regressor
$\bar{v}_{in}$	= normalized induced velocity
σ	= standard deviation

I. Introduction

Multi rotor drones are widely used currently as an efficient tool in multiple applications such as reconnaissance, package delivery, agriculture monitoring, filming and even personal transportation. Multi-rotor drones are equipped with individual rotors producing both propulsion and control power and frequently operate in non-hovering conditions in out-door environments. During flights with non-static incoming flow in these conditions, the aerodynamic characteristics of these rotors are different from those modeled in static conditions and considerable free-stream induced aerodynamic effects become apparent.

Drones are able to operate in conditions where additional aerodynamic effects occur without full knowledge of them due to the high update rates of sensors and robustness of the controller [1–3]. However, knowledge of these effects will be necessary for controllers capable of fully exploring the flight envelope [4, 5] such as high speed flights with aggressive maneuvers. Next to the controller enhancement, the modeling of these aerodynamic effects is also desirable of providing better attitude estimation [6] and refining the design process [7]. In addition, full knowledge of these aerodynamic effects is also required for high fidelity simulation platforms [8] and finally, global models need to be established for flight envelope computation [9], which is the main motivation for the current work.

The main subject of this research is the quadrotor, one of the simplest possible multi rotor drones. The aerodynamic effects acting on quadrotors can be summarized as the force variation and moment variation compared to that in the hovering condition without ground effect. Several discussions about these forces and moment variations are present in the literature.

Most literature sources focus on improving the thrust model. Refs. [10] , for instance, elaborate the cause of thrust variation during translational flight. The modeling process is mostly derived from helicopter aerodynamic theories. Refs. [4, 10, 11] use momentum theory to develop the model of relationships between thrust efficiency, flight speed, and the angle of attack. Thrust calculation of a single rotor according to blade element theory is adopted [12–15]. Momentum theory and blade element theory are also combined and a so-called blade element momentum (BEMT) theory is used to enhance the thrust model accuracy [16–18] .

Drag forces, which are mostly defined in the blade plane of multi-rotor drones, are also discussed in literature. The blade flapping effect is considered the main cause of drag force [10, 15, 19–21]. According to Ref. [12, 17, 19, 20], lift also induces an aerodynamic drag on the blade elements and generates a hub force perpendicular to the thrust. Besides the resistance caused by the rotor, the aerodynamic drag from the airframe is also considered [16], which is quadratic related to the flight speed.

Compared to forces, moment variations have received less attention in existing literature. The additional pitching moment due to the translational velocities is observed in the wind tunnel test presented in Ref. [22]. In the trim condition during forward flight, the aft rotors rotate faster than the front rotor. Damping effects [21], blade stiffness [11] and drag forces [21] are considered to cause these moments as well. The bare airframe itself may also generate a pitching

moment [8]. To the best of the author's knowledge no high fidelity models of the aerodynamic moments exist in the literature.

Besides the forces and moments generated from individual propellers and airframe, rotor-rotor and airframe-rotor interaction effects are suspected to greatly influence the aerodynamic forces and moments. Ref. [13] shows that the interaction between multiple rotors deteriorates the total thrust from wind tunnel tests. Ref. [8] divides the forces and moments into propulsion, airframe and interaction units, and the research of interaction terms is still ongoing. Models considering the interaction effect have been made [7][23] based on physical theory and engineering assumptions, however, not validated with in-flight data. The actual effects of these interactions on thrust, drag and moments remain largely unknown, which should be investigated with free flight experiments.

The main contribution of this research is further revealing the effect of above interactions from flight data and establish accurate force and moment models taking account of these effects. To this end, multiple free flight tests have been carried out. Based on the data from these tests, it is shown that the interaction effects deteriorate the well-established thrust and drag model based on the first principles, and in addition demonstrate significant inaccuracies in the pitch and rolling moment predictions obtained with the broadly accepted quadrotor hovering model, i.e the distance of propellers times their thrust differences. Furthermore, it is shown that the yawing moment is strongly influenced by the incoming flow during high speed flight, which has not been discussed before.

To establish a high fidelity model of forces and moments which is valid in a larger flight envelope, a system identification approach is used. Different from the first-principles modeling approach derived from helicopter aerodynamic theory, system identification methods are proper choice for modeling these complex interaction effects. Specifically, a gray-box model is established which combines the information from prior-physical knowledge of rotor-craft theory with experimental data obtained during high speed flight, and possesses both reliability of physical theory and accuracy of observation.

Several system identification techniques can be applied to establish nonlinear gray-box models, depending on the structure of the model, such as polynomial functions, multivariate spline-functions, neural networks, etc. Among them, a simple but effective piecewise polynomial structure is selected. A stepwise method is used for determining the model structure by selecting terms from a large set of candidate terms. This technique has been used in the past for full-scale aircraft system identification [24–26] but has never been seen used for determining the aerodynamic model of a quadrotor. The model structure candidates are determined from prior-knowledge of rotor-craft aerodynamic theories as well as preliminary assumptions. The stepwise method selects candidates into the model in a stepwise scheme according to their contributions to the current model.

The identified models are compared with state-of-art force models considering aerodynamic effects as well as moment models established in hovering conditions. The validation results reveal around 20% improvement in the accuracy of force model and more importantly, over 80% improvement in the moment model in terms of the residual

root mean square (RMS) in non-hovering conditions. Although these models are specific to the Bebop platform, the methodologies can be generalized to other multi-rotor platforms.

The flight experiments are carried out in the Open Jet Facility (OJF), a large scale wind tunnel with a 3 meters aperture operated by the Delft University of Technology, as shown in Fig. 1. In contrast to static wind tunnel tests, free flights are performed in the OJF in order to negate the disturbance effect of a force balance and more importantly, to take dynamic motions into account. The wind tunnel provides 2.5 m by 2.5 m by 5.0 m space to carry out these flights. A large number of different flight maneuvers are made to fully excite the system and a maximum air speed up to 14 m/s is achieved. An of-the-shelf quadrotor (Parrot Bebop) running open-source autopilot (Paparazzi) is used in these flights. The standard build-in inertia measurement unit (IMU) running at 512 Hz and external motion capture systems (Optitrack) running at 360 Hz are sensor-fused for data acquisition [27].



Fig. 1 Open Jet Facility (OJF), a large-scale wind tunnel, and the tested quadrotor.

A normalization method for modeling multi-rotor drones in terms of dimensionless coefficients is proposed in this research. The dimensionless aerodynamic coefficients and states are analogous to those used for single rotorcraft. Moment coefficients are for the first time introduced for quadrotor drones, taking into account their multi-rotor characteristics. The gray-box model established will be presented in dimensionless form as a mapping between dimensionless states and force (moment) coefficients. These dimensionless coefficients are also useful for revealing the interaction effects, and even for comparing the aerodynamic properties of different drone models.

The article is organized as follows. Chapter II introduces coordinate definition and hovering model definition. Chapter III depicts the stepwise method and provides definition of dimensionless aerodynamic coefficients and other dimensionless variables related to the model. Chapter IV describes the flight test for this research and discusses the interaction effects observed from flight data. The identification process and results can be found in Chapter V and VI respectively.

II. Preliminary Modeling

A benchmark model is introduced in this chapter with the aim of further introducing the gray-box model and make comparisons between them. First, the two coordinate systems, in the form of the ground frame and the body frame, are defined (Fig. 2). For the ground frame, x_G is defined towards the wind tunnel nozzle, in order words, into the free stream; z_G is aligned with the gravity direction pointing downwards. The body frame is fixed to the vehicle with the center of gravity at the origin. x_B is aligned with the nose direction, y_B points to the right and z_B points against the thrust direction.

Airspeed, the flight speed with respect to the air stream, is defined as

$$\mathbf{V} = \mathbf{V}_g - \mathbf{V}_{wind} \quad (1)$$

where $\mathbf{V}_g$ and $\mathbf{V}_{wind}$ indicate ground speed and wind speed respectively. The projection of airspeed on the body frame is expressed as $\mathbf{V} = [u \quad v \quad w]^T$. Angle of attack α and sideslip angle β are defined as

$$\alpha = \arcsin(w/V) \quad \beta = \arcsin(v/\sqrt{v^2 + u^2}) \quad (2)$$

where $V = \|\mathbf{V}\|$. Note that since the quadrotor is able to hover and reverse, these two angles are singular when airspeed equals zero.

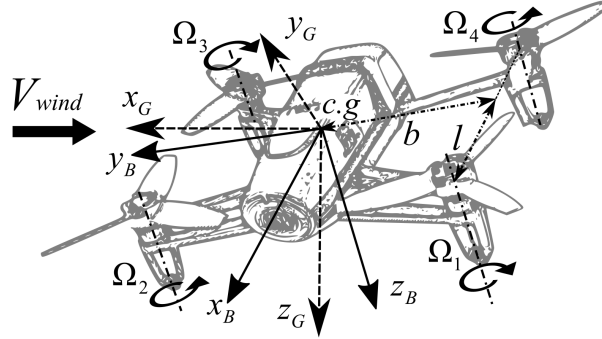


Fig. 2 Coordinate systems definition and sketch of Parrot Bebop quadrotor.

Rotor speeds (in rad/s) are expressed as $[\Omega_1 \quad \Omega_2 \quad \Omega_3 \quad \Omega_4]$ respectively. Fig. 2 shows the rotor index and their rotation directions of Parrot Bebop quadrotor, which is the object to be modeled in this paper.

For simplicity, the quadrotor is regarded as a rigid-body of which the translational and rotational dynamic equations can be written as

$$\dot{\mathbf{V}} + \boldsymbol{\Omega} \times \mathbf{V} = \mathbf{R}_{BG}\mathbf{g} + \mathbf{F}/m \quad (3)$$

$$\mathbf{I}_v \dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times \mathbf{I}_v \boldsymbol{\Omega} = \mathbf{M} + \mathbf{M}_{I_p} \quad (4)$$

where $\mathbf{g} = [0 \ 0 \ g]^T$ indicates the gravity vector expressed in the ground frame. $\boldsymbol{\Omega} = [p \ q \ r]^T$ represents the angular velocity expressed in the body frame. The aerodynamic forces and moments are denoted as $\mathbf{F}$ and $\mathbf{M}$ respectively, which are expressed in the body frame as well. $\mathbf{R}_{BG}$ is the rotational matrix from the ground frame to the body frame. $\mathbf{I}_v$ stands for the inertia matrix and m indicates the mass of the vehicle. $\mathbf{M}_{I_p}$ represents the moments due to gyroscopic effects and rotor spin-up torque; the later has been found significantly influence the Bebop quadrotor [28]

$$\mathbf{M}_{I_p} = \begin{bmatrix} qI_p(-\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4) \\ pI_p(\Omega_1 - \Omega_2 + \Omega_3 - \Omega_4) \\ I_p(-\dot{\Omega}_1 + \dot{\Omega}_2 - \dot{\Omega}_3 + \dot{\Omega}_4) \end{bmatrix} \quad (5)$$

The positioning of a quadrotor in 3D space is controlled by changing its attitude and total thrust. The attitude can be changed by differential thrust. Specifically, rotor speed difference between front and aft rotors produces a pitching moment, while a rolling moment can be produced by differential thrust between left and right rotors. The rotor reaction torque is used to generate a yawing moment which the quadrotor uses to control its heading.

The emphasis of modeling is on the aerodynamic force vector $\mathbf{F}$ and the aerodynamic moment vector $\mathbf{M}$. Before establishing a gray-box model for $\mathbf{F}$ and $\mathbf{M}$, a hovering model that is only valid in hovering condition, is introduced as a benchmark for comparison

$$\mathbf{F}_h = \begin{bmatrix} 0 \\ 0 \\ -\kappa_0 \sum \Omega_i^2 \end{bmatrix} \quad (6)$$

$$\mathbf{M}_h = \begin{bmatrix} b\kappa_0(\Omega_1^2 + \Omega_2^2 - \Omega_3^2 - \Omega_4^2) \\ l\kappa_0(\Omega_1^2 - \Omega_2^2 - \Omega_3^2 + \Omega_4^2) \\ \tau_0(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2) + \lambda_r r \end{bmatrix} \quad (7)$$

where l and b are geometry parameters of the quadrotor as Fig. 2 shows. κ_0 , τ_0 and λ_r are constant coefficients. Note that the aerodynamic forces and moments are expressed in the body frame in this paper. Therefore, the third component of $\mathbf{F}_h$ equals to the negative of the total thrust. Meanwhile, the first two components of $\mathbf{F}_h$ equal zero, which means that rotor in-plane forces (i.e drag forces) are neglected.

However, significant in-plane drag has been found [6, 12, 17, 19] and thrust also varies with the flight speed beyond the hovering regime [11]. Furthermore, aerodynamic moments are found to be completely different from what the

hovering model predicts. These studies indicate the importance of finding a model which is valid in a larger flight envelope.

In this paper, a gray-box model is identified from high-speed free-flight data obtained in a wind tunnel. The aerodynamic effect of individual rotors, the rotor-rotor and rotor-airframe aerodynamic interactions in high-speed conditions are considered in this gray-box model as well.

III. Methodologies

A. Nondimensionalization

Dimensionless aerodynamic coefficients are convenient for comparisons between different conditions and platforms. For a single rotor, forces and moments can be normalized by rotor speed and reference area [8]. However, for multi-rotor aircraft such as quadrotors, determining the aerodynamic coefficient of each rotor could be impracticable using system identification approach since only joint forces are measurable by the 3-axis accelerometer located at the center of gravity. Furthermore, the local airspeed differs between rotors because of complex aerodynamic interactions, which makes the rotor-by-rotor modeling approach impractical.

In this research, a novel nondimensionalization approach is proposed which is based on an assumption that aerodynamic forces and moments are mainly generated by the rotor system. A geometric average of rotor speeds is used to represent the effect of multi-rotors

$$\bar{\Omega} = \sqrt{\frac{\sum_{i=1}^N \Omega_i^2}{N}} \quad (8)$$

where N is the number of rotors and equals 4 for quadrotor. In most cases, rotors are the same size with radius R . Afterwards, forces and moments acting on the entire vehicle can be normalized by the average rotor speed

$$C_z = \frac{F_z}{\rho(N\pi R^2)(R\bar{\Omega})^2}, \quad C_x = \frac{F_x}{\rho(N\pi R^2)(R\bar{\Omega})^2}, \quad C_y = \frac{F_y}{\rho(N\pi R^2)(R\bar{\Omega})^2} \quad (9)$$

$$C_l = \frac{M_x}{\rho b(N\pi R^2)(R\bar{\Omega})^2}, \quad C_m = \frac{M_y}{\rho b(N\pi R^2)(R\bar{\Omega})^2}, \quad C_n = \frac{M_z}{\rho b(N\pi R^2)(R\bar{\Omega})^2} \quad (10)$$

where b is the reference length chosen arbitrarily as long as it represents the geometry size of a specific vehicle. Since F_z is opposite to thrust which brings intuitive inconvenience, $T = -F_z$ is used as the total thrust force and $C_t = -C_z$ as the thrust coefficient. Note that T is interpreted as the joint of rotor thrust and drag force along the body vertical axis.

Translational and angular velocities, which have been found to significantly influence the abovementioned coefficients [10, 19], are normalized by

$$\mu_x = \frac{u}{\bar{\Omega}R}, \quad \mu_y = \frac{v}{\bar{\Omega}R}, \quad \mu_z = \frac{w}{\bar{\Omega}R} \quad (11)$$

$$\bar{p} = \frac{pb}{\bar{\Omega}R}, \quad \bar{q} = \frac{qb}{\bar{\Omega}R}, \quad \bar{r} = \frac{rb}{\bar{\Omega}R} \quad (12)$$

The horizontal component of the advance ratio $\mu = \sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2}$, which is defined as $\mu_h = \sqrt{\mu_x^2 + \mu_y^2}$, is used to analyze interaction effects in this research. These dimensionless parameters are analogous to those used for single-rotor aircraft.

The rotor speeds are normalized by

$$\omega_i = \frac{\Omega_i}{\bar{\Omega}} \quad (13)$$

It is assumed that $\bar{\Omega} > 0$ always holds to avoid the singularity, because the status that all rotors are stopped is out of the scope of this paper.

The moments for controlling attitude are produced by differential thrust from rotor speeds differences. Here three normalized inputs for roll, pitch and yaw controls are defined

$$u_p = (\omega_1^2 + \omega_4^2) - (\omega_2^2 + \omega_3^2) \quad (14)$$

$$u_q = (\omega_1^2 + \omega_2^2) - (\omega_3^2 + \omega_4^2) \quad (15)$$

$$u_r = -(\omega_1^2 + \omega_3^2) + (\omega_2^2 + \omega_4^2) \quad (16)$$

where signs and numbers are in accordance with the definition in Fig. 2, which may change for different types of quadrotor vehicles.

B. Stepwise System Identification

This section introduces the system identification approach applied to establish the gray-box model. Specifically, a regression method together with a model structure selection algorithm is used for determining mappings from dimensionless states to aerodynamic force and moment coefficients. The relation of model outputs to measurements satisfies

$$\mathbf{z} = \mathbf{y} + \boldsymbol{\epsilon} = \mathbf{A}\boldsymbol{\theta} + \boldsymbol{\epsilon} \quad (17)$$

where $\mathbf{z} \in \mathbb{R}^N$ stands for N measured dimensionless forces and moments. $\mathbf{y} = \mathbf{A}\boldsymbol{\theta}$ denotes model output and $\boldsymbol{\epsilon} \in \mathbb{R}^N$ vector indicates model residuals. $\mathbf{A} \in \mathbb{R}^{N \times p}$ is the regressor matrix with each column as a regressor which is an arbitrarily combination of independent variables. $\boldsymbol{\theta} \in \mathbb{R}^p$ stands for parameters of regressors to be estimated using , e.g Original Least Square (OLS) estimator

$$\hat{\boldsymbol{\theta}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{z} \quad (18)$$

where $\hat{\theta}$ is the optimal parameter estimation that minimizes the sum of squares of the residual ϵ . The process of model structure selection is concerned with the choice of particular regressors in the A matrix.

In this paper, two steps are taken in the model structure selection. The first step is defining a candidate regressors set using prior knowledge; the second step is selecting candidates using a selection algorithm.

A method to rigorously define candidate sets is introduced. Suppose $y(x_1, x_2, x_3)$ is a model with three independent variables and unknown model structure, the candidate set of y can be denoted by $\mathcal{S}(y)$. In this research, the model structures are in the form of polynomial functions. Now denote the basis of a d th order polynomial function of $\mathbf{x} = (x_1, x_2, x_3)$ as $P^d(\mathbf{x})$ and then the candidate set consisting of arbitrary polynomial terms can be defined. For example, if $\mathcal{S}(y) = \{P^2(x_1, x_2), P^2(x_1, x_2)x_3\}$, the candidate set of y contains regressors from $P^2(x_1, x_2) = \{x_1, x_2, x_3, x_1^2, x_2^2, x_1x_2\}$ and from $P^2(x_1, x_2)x_3 = \{x_1x_3, x_2x_3, x_3x_3, x_1^2x_3, x_2^2x_3, x_1x_2x_3\}$. Define the multiplication of two sets as a set containing non-repetitive products of elements from the two sets

$$\{a_1, a_2, \dots, a_m\}\{b_1, b_2, \dots, b_n\} = \{a_1b_1, \dots, a_1b_n, a_2b_1, \dots, a_2b_n, \dots, a_mb_1, \dots, a_mb_n\} \quad (19)$$

Then $\mathcal{S}(y)$ can be expressed in a simplified form according to the law of association

$$\mathcal{S}(y) = \{P^2(x_1, x_2), P^2(x_1, x_2)x_3\} = \{P^2(x_1, x_2)\{1, x_3\}\} \quad (20)$$

A general formulation of $P^d(\mathbf{x})$ is

$$p^d(x_1, x_2, \dots, x_n) = \left\{ \prod_{i=1}^n x_i^{k_i} \mid 0 \leq \sum_{i=1}^n k_i \leq d, k_i \in \{0, 1, 2, \dots, d\} \right\} \quad (21)$$

of which the total number of elements can be calculate by

$$\hat{d} = \frac{(d+n)!}{n!d!} \quad (22)$$

After determining the candidate set, a so-called (forward-backward) stepwise regression algorithm is applied to select regressors to build the model. The algorithm is summarized in the Appendix. Readers may refer [25] for more details.

IV. Data acquisition and analysis

A. Experimental Setup

In order to identify force and moment models in high-speed flight regimes, free flights are performed in a large-scale wind tunnel for data acquisition. The tested quadrotor is Parrot Bebop without bumper, as Fig. 2 shows. The native

autopilot of this off-the-shelf drone is replaced by Paparazzi [29], an open-source autopilot which runs at 512 Hz, and which is capable of performing aggressive maneuvers. Incremental Nonlinear Dynamic Inversion (INDI) guidance law and attitude controller have been programmed in Paparazzi [3, 28] to guarantee the position tracking performance against strong wind, which is essential for flight tests in our research. The quadrotor is equipped with a closed-loop Brushless DC motor controller; an MPU6050 Inertia Measurement Unit (IMU) including a 3-axis accelerometer and gyroscope. The inertia of Bebop is measured using the approach introduced in Ref. [30] with a percent error less than 5%. These parameters of tested quadrotor are listed in the Table 1

Flight tests are performed in the Open Jet Facility (OJF), a large-scale wind tunnel, operated by Delft University of Technology. The drone is controlled to maneuver in a confined area that is approximately 5.0 m long, 2.5 m wide and 2.5 m high. Wind speed is varied from 0 to 14 m/s with 2 m/s intervals to simulate flights at different airspeeds. An external motion capture system is applied to measure velocities and positions of the drone for indoor navigation. As Fig. 3 shows, five waypoints in the flight area are set to conduct flight maneuvers. To perform longitudinal maneuvers, the quadrotor can be controlled to track point A and B alternately. Similarly, waypoints C and D are set for performing lateral maneuver. To perform vertical maneuvers during forward flight, the drone is controlled to stay at point O and climb or descend with 2 m/s. Longitudinal, lateral and vertical maneuvers are conducted at varying heading angles denoted by ψ which are defined in Fig. 3; the heading angle is increased from 0 to 360° in steps of 45°. To identify the yawing moment model considering aerodynamic effects, yaw maneuvers are carried out at point O by changing ψ in steps of 45° both clockwise and counter-clockwise.

B. Data Preprocessing

Flight data for system identification are collected by on-board and external sensors. Specifically, rotor speeds are observed by the motor controller, angular rates and specific forces are measured by gyroscope and accelerometer respectively. These measurements are logged on-board at 512 Hz. The external motion capturing system (10 × OptiTrack Prime 17 W cameras) measured the position of 6 markers fixed on the vehicle at 360 Hz, with a standard deviation less than 0.2 mm. Henceforth the quadrotor position, attitude and ground speed are derived from these marker positions and re-sampled to 512 Hz to align with on-board measurements.

Measurements from the two sources have been fused using an Extended Kalman Filter (EKF) with the aim of calculating IMU bias [27]. The unbiased IMU measurements are further filtered by a 4th-order Butterworth lowpass filter. Power spectral density (PSD) of accelerometer and gyroscope measurements from one flight are plotted in Fig. 4.

Table 1 Inertia and geometric parameters of Parrot Bebop

m [kg]	I_x [kg · m ²]	I_y [kg · m ²]	I_z [kg · m ²]	I_{xz} [kg · m ²]	I_p [kg · m ²]	b [m]	l [m]	R [m]
0.389	0.000906	0.001242	0.002054	1.42E-05	3.39E-06	0.0775	0.0975	0.064

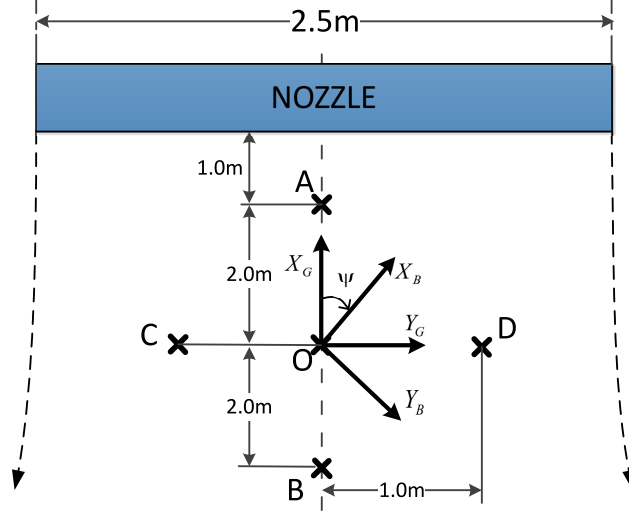


Fig. 3 Top view diagram of flight maneuvers performed in the wind tunnel. Forward and backward flights are conducted by tracking point A and B by turns. Lateral flights are performed between C and D. Descend and ascend flights, yaw maneuvers are made at point O. The heading angle ψ is defined as the angle between x_B and x_G (clockwise positive).

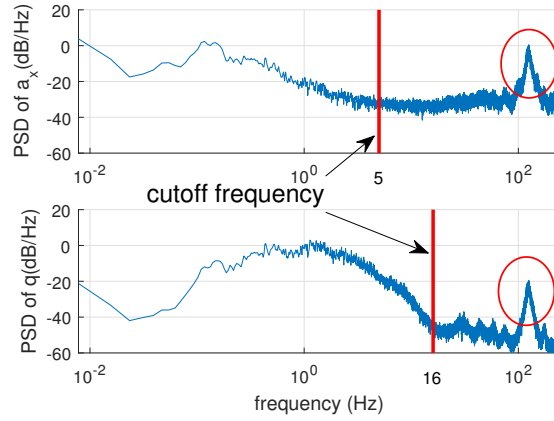


Fig. 4 Power spectrum density (PSD) of accelerometer measurement and gyroscope measurement (acc_x and q) by Welch's method in Matlab. Cut-off frequencies are chosen as 5 Hz and 16 Hz respectively. Circled parts indicate the noise caused by an unbalanced rotor which is inevitable.

There is a resonance peak at around 120 Hz which is most likely caused by rotor imbalance (rotors rotate at around 7000 RPM = 117 Hz). Filter cut-off frequencies are chosen as 5 Hz and 16 Hz respectively, leading to a considerable noise reduction as shown in Fig. 5.

Finally, force and moment measurements are derived from the processed IMU data. The specific force times the mass of quadrotor equals the resultant non-gravitational force, namely $\mathbf{F}$. The moment $\mathbf{M}$ can be obtained from Eq. (4) where angular velocity $\mathbf{\Omega}$ is obtained from the processed gyroscope measurement.

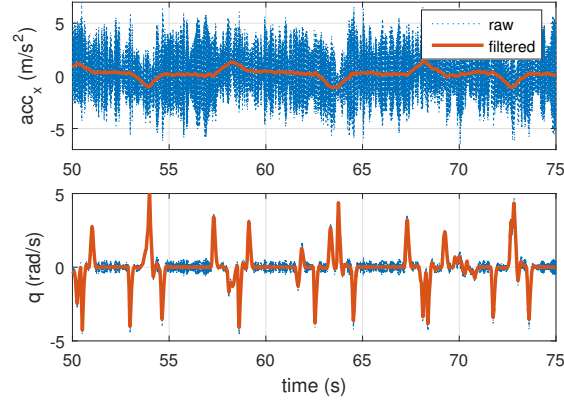


Fig. 5 Comparison between raw and filtered measurements from accelerometer and gyroscope.

C. Complex Aerodynamic Effect

As outlined in the introduction, complex aerodynamic effects such as interactions between quadrotor components, have been clearly observed from the flight test data. These effects can significantly affect external forces and moments and has to be taken into account when creating a high fidelity model.

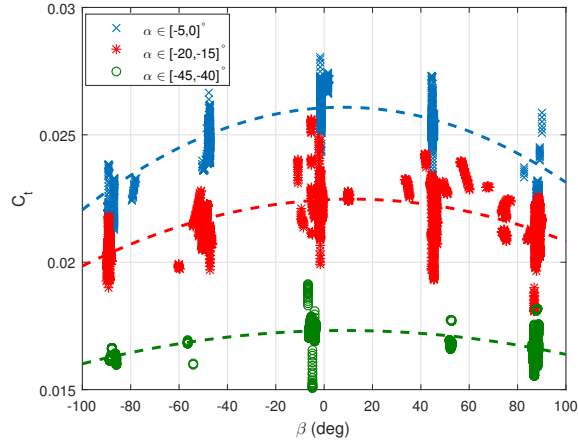


Fig. 6 C_t vs. sideslip angle in different angle of attack.

Thrust coefficients C_t and corresponding sideslip angles β are shown in Fig. 6. These data are divided into three groups according to angles of attack α . The intervals of α are presented in the figure as well, while the interval of advance ratio is set as $\mu \in [0.1, 0.11]$ ($V \approx 5$ m/s). Note that only data within these intervals are plotted. Trend lines are also given for a better illustration of the C_t variation in different β . Apart from the vertical shift of C_t , the effect of β on C_t also varies with α . When α is negative with large absolute value, the mean value of C_t seems uncorrelated to β . As α is approaching zero, data with larger $|\beta|$ have smaller C_t , which could be interpreted as the thrust degradation caused by disturbance from the fuselage of Bebop quadrotor in high sideslip flights when the aft rotors are within the fuselage wake.

In contrast, when α is decreased and becomes sufficient small, the aft rotors are outside the fuselage wake, therefore the sideslip angle does not influence the thrust coefficient. During the experiment, a clear shrill sound was produced while the aft rotors were obstructed by the fuselage, which can be regarded as another evidence that supports this hypothesis.

Fig. 7 presents the horizontal force coefficient defined as $C_h = \sqrt{C_x^2 + C_y^2}$ which indicates the total aerodynamic resistance projected on the x_B - y_B plane. The angle of attack is selected around -20° which is typical during forward flight. The distribution of data points with small sideslip angle ($|\beta| \in [0, 10]^\circ$) shows an almost linear relationship between C_h and horizontal advance ratio μ_h . As sideslip angle grows, the drag coefficient also increases, and when $|\beta| \in [80, 90]^\circ$ a quadratic tendency appears. This may be due to the fact that the fuselage of the Bebop has larger a projection area on y_B direction, as Fig. 2 shows.

Besides the sideslip angle, the angle of attack also affects C_h . Since a large portion of the drag is produced by the rotors, thrust degradation caused by aerodynamic interactions can lead to induced drag reduction and subsequently decrease the total drag force. This drag reduction was not only observed in large sideslip angle but also straight forward flights when fuselage wake does not influence rotors. As can be seen in Fig. 8 where the sideslip angle is small ($\beta \in [-10, 10]^\circ$), however C_h is found reduced when α is close to or above zero. This might be explained by the fact that the front rotors can obstruct the aft rotors and degraded their aerodynamic characteristics as well. Note that this drag force reduction is not obvious in the low-speed region since the flight speed is not high enough for this interaction. The drag force variance due to interactions should be considered as well in a high fidelity model.

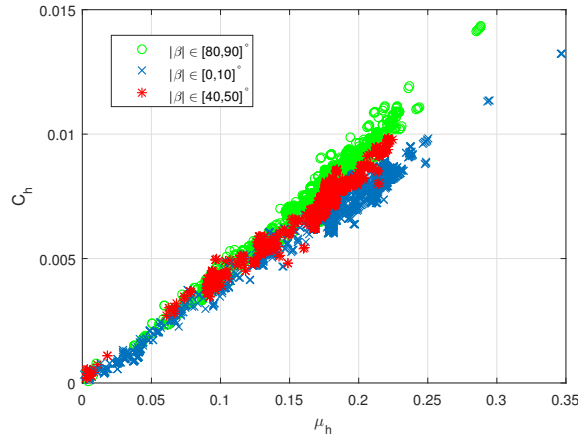


Fig. 7 C_h vs. μ_h in different sideslip angle intervals. Plot shows that horizontal force coefficient varies with respect to sideslip angle.

Fig. 9 presents the pitching moment coefficient C_m versus the advance ratio μ at different sideslip angles. Significant differences can be found in different β intervals. For these data, pitch control u_q are chosen close to zero ($u_q \in [-0.05, 0]$) which means that almost no pitching moment is present in hovering condition because the front and aft rotors are nearly at the same speed. It is evident that the pitching moment increases significantly as the flight speed grows when

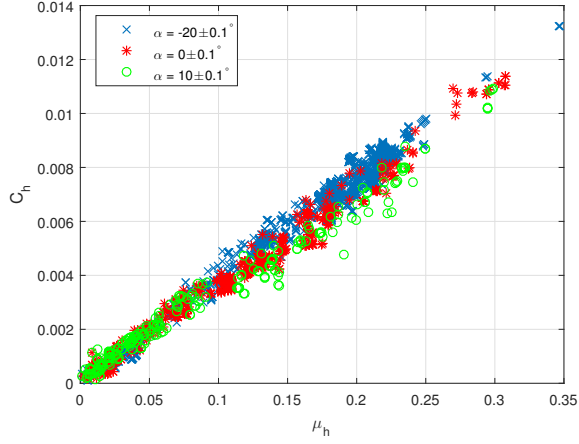


Fig. 8 C_h vs. μ_h in different angles of attack when sideslip angle $\beta \in [-10, 10]^\circ$. A larger angle of attack can reduce the horizontal force coefficient.

$|\beta| \in [-10, 10]^\circ$ compared to other sideslip angles. This pitch-up moment might be caused by blade flexibility [11] or interactions between rotors.

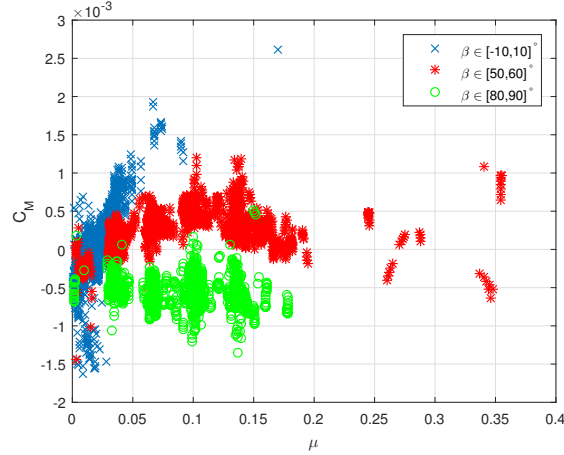


Fig. 9 C_m in different sideslip angles when $u_q \in [-0.05, 0]$. Large nose up moment can be observed when sideslip angle is small ($\beta \in [-10, 10]^\circ$).

The angle of attack is found to be positively related to the pitching moment during forward flight. Fig. 10 shows the data and trend lines of C_m versus α with $u_q \in [-0.01, 0.01]$ and $\beta \in [-10, 10]^\circ$. In these cases, the aft rotors and front rotors have almost the same rotor speeds. In general, C_m is positively correlated with the angle of attack indicating the longitudinal instability of a quadrotor. The slope is larger with a higher advance ratio, which is consistent with results given by Fig. 9. More importantly, the C_m is almost always positive, indicating that a nose up aerodynamic moment appears in quadrotor forward flight even with a large negative angle of attack. This coincides with the result from Ref. [22] that the aft rotors need to rotate much faster than the front rotors in trim conditions. Similarly, for instance, flying to the right (with $\beta = 90^\circ$) can produce a negative rolling moment which requires the left rotors to rotate faster in

order to keep balance. This phenomenon is strongly present in the data; simple hovering models that neglect this effect will produce highly inaccurate moment predictions in fast flight regions.

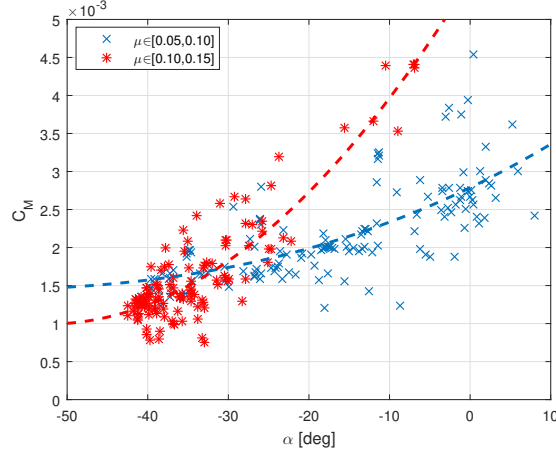


Fig. 10 C_m vs. α in different advance ratio intervals with $u_q \in [-0.01, 0.01]$ and $\beta \in [-10, 10]^\circ$.

The yawing moment is also found to be influenced by the sideslip angle. Notate the difference between the measured yawing moment and that calculated by the hovering model by ΔM_z , which is regarded as the additional yawing moment due to the aerodynamic effects.

$$\Delta M_z = M_z - [\tau_0(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_3^2) + \lambda_r r] \quad (23)$$

As is shown in Fig. 11, ΔM_z is negatively related to the sideslip angle in general. When $|\beta| < 40^\circ$, ΔM_z is in the vicinity of zero. However its dispersion suddenly increases when $|\beta| > 40^\circ$. At the same time, one aft rotor starts to be obstructed by the fuselage. The airframe-rotor aerodynamic interaction might occur in this situation as the cause of the sudden increased yawing moment.

V. Quadrotor Model Structure Candidates

This section introduces the determination of candidate structure sets for the quadrotor dimensionless aerodynamic force and moment models, based on prior physical knowledge and observations. Then, the stepwise regression algorithm can be carried out to obtain the final model structure, which will be presented in a subsequent section.

A. Force Model Candidates

F_x and F_y denote forces perpendicular to the thrust direction brought by aerodynamic resistance, of which the lift-induced drag and the blade flapping effect are the two major causes [6, 12, 19]. A widely accepted drag model of a

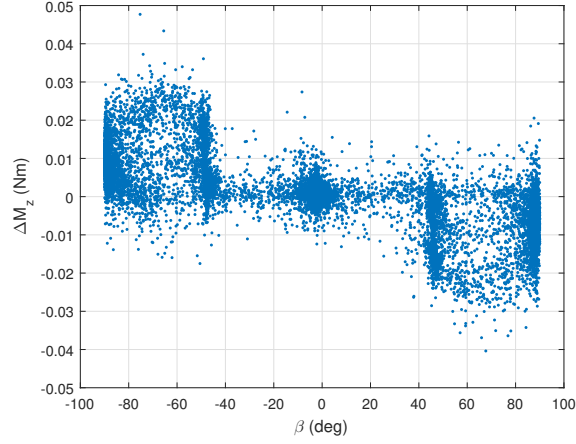


Fig. 11 Additional aerodynamic yawing moment ΔM_z with respect to the hovering model. ΔM_z is negative related to the sideslip angle indicating that an aerodynamic moment related to β exists.

single rotor is [6, 12]

$$F_{x,i} \propto u_i \Omega_i \quad (24)$$

where $F_{x,i}$ and u_i stand for the in-plane force and local velocity of the i_{th} rotor on the x_b direction respectively. In addition, as section C shows, airframe affects the drag force as well, especially at large sideslip angles. Therefore, the square of the velocity has been added into the drag model [16]. The drag force on x_B direction can thus be expressed as

$$F_x = \kappa_{d,1} \sum_{i=1}^4 \Omega_i u_i + \kappa_{d,2} u^2 \quad (25)$$

where $\kappa_{d,1}$ and $\kappa_{d,2}$ are constants. Recall Eq. (8911), the normalized form of Eq. (25) can be obtained

$$C_x = C_{x,1} \mu_x + C_{x,2} \mu_x^2 \quad (26)$$

note that the rotor speed term $\sum_{i=1}^4 \Omega_i$ in Eq. (25) disappears after normalization. This process needs to replace the arithmetic mean of rotor speeds by their geometrical mean, which does not lose accuracy in most flight conditions (0.8% relative error on average).

The above model can produce accurate predictions in most low-speed cases. However, it loses accuracy when interaction effects appear. For comparison with the gray-box model established in this research, the drag model in Eq. (26) is named the *reduced model*.

An additional term $C_{x,2,(\mu_x, \mu_y, \mu_z)}$, of which the exact structure is to be determined, is added to the reduced model. The variables in the subscript parentheses indicate the independent variables of $C_{x,2}$. A preliminary model structure of

C_x can be

$$C_x = C_{x,1}\mu_x + C_{x,2}(\mu_x, |\mu_y|, \mu_z) \quad (27)$$

Note that the $C_{x,2}\mu_x^2$ term is moved into the second term since it has a negligible effect in the flight regime as the data shows.

Recall the observation in the Section. C; β , α and μ_h greatly influence the drag coefficient. Singularities in α and β , however, could occur in hovering conditions, therefore three components of the advance ratio are chosen as the independent variables. Absolute values of μ_y are used due to the symmetry of the quadrotor.

Similarly, for C_y we have

$$C_y = C_{y,1}\mu_y + C_{y,2}(|\mu_x|, \mu_y, \mu_z) \quad (28)$$

The stepwise regression algorithm can be applied to determine the exact structure of the unknown parts of the above models. The candidate sets of C_x and C_y are chosen as

$$S(C_x) = \{P^3(\mu_x, |\mu_y|, \mu_z)\} \quad (29)$$

$$S(C_y) = \{P^3(|\mu_x|, \mu_y, \mu_z)\} \quad (30)$$

The terms μ_x and μ_y are always the first candidates to be tested by the model structure selection algorithm when assembling respectively the models for C_x and C_y .

F_z is derived by taking account of the thrust variation. In general, the thrust of the i th rotor with constant pitching angle can be expressed as [31]

$$T_i = \frac{\rho a B c \omega_i^2 R^3}{2} \left(\frac{\theta_r}{3} + \frac{(u_i^2 + v_i^2)\theta_r}{2\omega_i^2 R^2} + \frac{-w_i + v_{in,i}}{2\omega_i R} \right) \quad (31)$$

where a is the lift curve slope of the blade profile, B represents the number of blades, c is the blade chord length and θ_r stands for the rotor pitch angle. These parameters are related to rotor design and for quadrotors; they normally are all constants. $v_{in,i}$ indicates the induced velocity of the i th rotor.

A common way to model F_z is projecting the thrust of four rotors on the body frame and assuming that local velocities and induced velocities of the rotors are identical, yielding

$$F_z = - \sum_{i=1}^4 T_i = \kappa_{t,1} \sum_{i=1}^4 \Omega_i^2 + \kappa_{t,2}(u^2 + v^2) + \kappa_{t,3}(-w + v_{in}) \sum_{i=1}^4 \Omega_i \quad (32)$$

where $\kappa_{t,1}$, $\kappa_{t,2}$ and $\kappa_{t,3}$ are constants. Eq. (32) is the model structure adopted in several researches [13, 31]; it is indicated as the *reduced model* to compare against the new gray-box model. A dimensionless form of Eq. (32) can be

Table 2 Procedure of identifying the reduced model and the gray-box model

	Reduced model	Gray-box model
step 1	Data acquisition	
step 2	Determined structure	Define structure candidates $\mathcal{S}(y)$
step 3	Parameter estimation	Stepwise regression
	Final model	

calculated by substituting Eq. (8,9,11) into Eq. (32), yielding

$$C_z = C_{z,0} + C_{z,1}(\mu_x^2 + \mu_y^2) + C_{z,2}(-\mu_z + \bar{v}_{in}) \quad (33)$$

where $C_{z,0}$, $C_{z,1}$ and $C_{z,2}$ are constants. $\bar{v}_{in}$ indicates the dimensionless induced velocity normalized by $\bar{\Omega}R$. The induced velocity $\bar{v}_{in}$ can be calculated by [4]

$$\bar{v}_{in} = \frac{C_{t,h}}{2\sqrt{\mu_x^2 + \mu_y^2 + (-\mu_z + \bar{v}_{in})^2}} \quad (34)$$

where $C_{t,h}$ is the thrust coefficient in the hover case which can be estimated accurately by conducting hovering flight experiments.

As mentioned earlier, interaction effects could degrade rotor thrust. This effect is not considered in the reduced model (32), not to mention other unknown complex aerodynamic effects and drag force on the z_B direction. The deviation of C_z from these effects is denoted by $C_{z,3}$. Flight speed, the difference between aft and front rotors, vehicle angular velocities could be the individual variables of this unknown part. Thus a gray-box model of C_z can be formalized as

$$C_z = C_{z,0} + C_{z,1}(\mu_x^2 + \mu_y^2) + C_{z,2}(-\mu_z + \bar{v}_{in}) + C_{z,3}(|\mu_x|, |\mu_y|, \mu_z, |u_p|, |u_q|, |u_r|, |\bar{p}|, |\bar{q}|, |\bar{r}|) \quad (35)$$

Henceforth the stepwise regression algorithm is applied to determine the structure of C_z , of which the candidate structure set is chosen as

$$\mathcal{S}(C_z) = \{P^4(|\mu_x|, |\mu_y|, \mu_z)\{1, |\bar{p}|, |\bar{q}|, |\bar{r}|, |u_p|, |u_q|, |u_r|\}\} \quad (36)$$

Note that $(\mu_x^2 + \mu_y^2)$ and $(-\mu_z + \bar{v}_{in})$ are regarded as fixed regressors that have been added in the model before the selection.

Both the reduced model and the gray-box model can be established using the system identification method. Table 2 briefly compares the procedures of establishing these models.

B. Moment Model Candidates

Compared to the hovering case, additional aerodynamic moments can be produced during high-speed flight. First, the pitching moment could be a result of rotor resilience and the blade flapping effect [11]. Secondly, the vertical distance between rotor planes and the center of gravity also brings moments due to rotor drag [21]. Thirdly, angular rate (dynamic) damping term also contributes to the total aerodynamic moment [21]. Fourthly, as was observed from the flight test data, aerodynamic interactions may degrade the thrust of the aft rotors and lead to additional pitch up moments. All the above factors are related to advance ratio and angular rates.

Taking account of these possible effects, a lumped model of the pitching moment coefficient C_m could be expressed as

$$C_m = C_{m,(\mu_x, \mu_y, \mu_z, \bar{q}, u_q)} \quad (37)$$

with advance ratio, dimensionless pitch rate and pitch input as independent variables.

Above pitch model neglects the influence of lateral variables u_p , u_r , p and r , which are of less effect on the pitching moment based on the flight data. However, μ_y is included in the model to handle the sideslip effect. To further simplify the candidate set, u_q and $\bar{q}$ are assumed linearly related to the pitching moment. Thus the lumped model (37) can be expressed as

$$C_m = C_{m,0,(\mu_x, |\mu_y|, \mu_z)} + C_{m,u_q,(\mu_x, |\mu_y|, \mu_z)} u_q + C_{m,\bar{q},(\mu_x, |\mu_y|, \mu_z)} \bar{q} \quad (38)$$

Based on the structure in Eq. (38), the candidate set of C_m is chosen as

$$\mathcal{S}(C_m) = \{P^5(\mu_x, \mu_z)P^2(|\mu_y|)\{1, \bar{q}, u_q\}\} \quad (39)$$

The preliminary structure of the rolling moment model can be similarly determined as

$$C_l = C_{l,0,(\mu_x, |\mu_y|, \mu_z)} + C_{l,u_p,(\mu_x, |\mu_y|, \mu_z)} u_p + C_{l,\bar{p},(\mu_x, |\mu_y|, \mu_z)} \bar{p} \quad (40)$$

of which the regressors are selected from the candidate set

$$\mathcal{S}(C_l) = \{P^5(\mu_y, \mu_z)P^2(|\mu_x|)\{1, \bar{p}, u_p\}\} \quad (41)$$

The yawing moment model is found to be much more complicated and u_r may not be linear to the model. Thus a preliminary structure is determined as

$$C_n = C_{n,(\mu_x, \mu_y, \mu_z, \bar{r}, u_r)} \quad (42)$$

of which the candidate set is chosen as

$$\mathcal{S}(C_n) = \{P^5(\mu_x, \mu_y, \mu_z)P^3(\bar{r})P^3(u_r)\} \quad (43)$$

VI. Results

A. Model Estimation Results

After defining candidate structure sets, all force and moment models are determined by the stepwise regression algorithm. This section provides the estimation result of C_z since thrust is the biggest concern in most modeling tasks. Due to limited space, other models are provided in the Appendix.

The reduced model is found to be accurate in low-speed regions since the interaction effects are weak. Thus C_z is estimated with structure (33) when $\mu < 0.05$ (approximately $V < 2$ m/s). The result at this low-speed regime is listed in Table 3.

For the flight regime where $\mu > 0.05$, the complex aerodynamic effects become apparent and the gray-box model of C_z is established. The model structures and parameters are listed in Table 4. The state space is equally divided into three partitions according to the sideslip angle. On each partition, a gray-box model of C_z has been identified. The first column of the table lists the model structure which is ranked by the order of selection. The second column gives the values of corresponding parameters. The third column provides the decreasing normalized-root-mean-square (NRMS) of the model residual after the corresponding regressor is added into the model. In general, regressors at the top are the most significant, with significance becoming less moving towards the bottom of the list.

Fig. 12, 13 present the residuals of gray-box models on the entire estimation data sets. $1 - \sigma$ intervals are given as well. In general, the residuals are confined to the interval. It also indicates that the gray-box model provides unbiased estimations because all residuals have a mean value closed to zero. Above properties demonstrate the validity of the model structure selected by the stepwise regression algorithm.

Table 3 Estimation results of C_z model, $\mu \leq 0.05$

reg.	$\hat{\theta}$
1	5.020E-02
$\mu_x^2 + \mu_y^2$	1.112E-01
$(\bar{v}_{in} - \mu_z)^2$	-9.420E-02
R2	0.953
NRMS	0.021

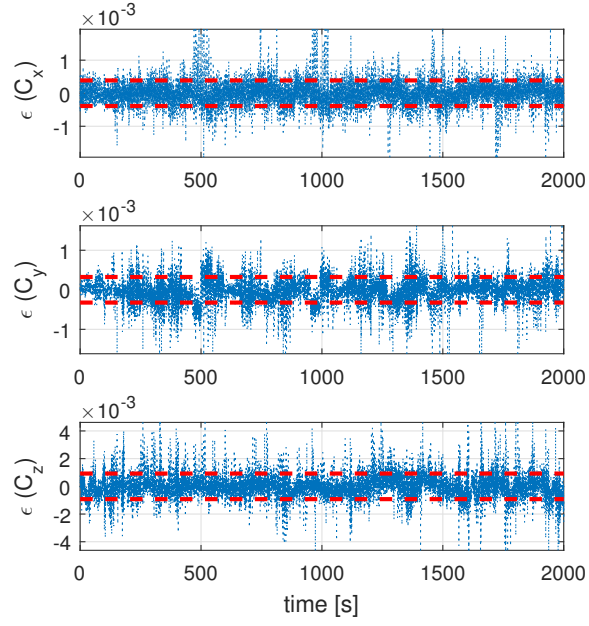


Fig. 12 Residuals of force models compared with $1\text{-}\sigma$ interval.

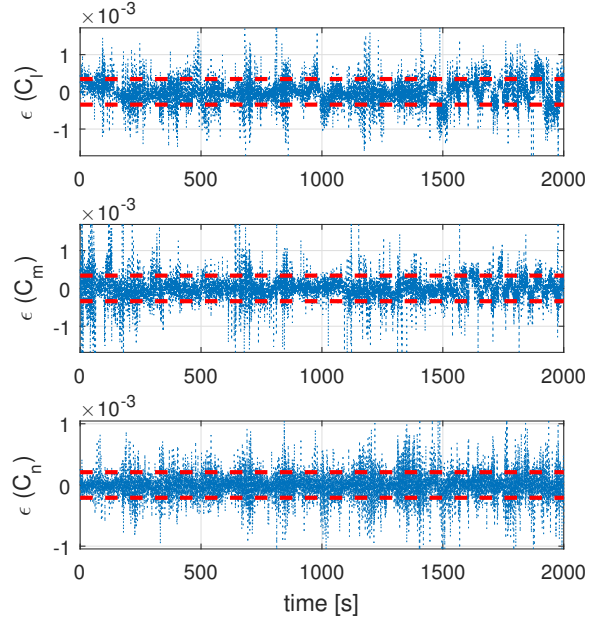


Fig. 13 Residuals of moment models compared with $1\text{-}\sigma$ interval.

Table 4 Estimations result of C_z model, $\mu > 0.05$

$ \beta \in [0, 30]$			$ \beta \in [30, 60]$			$ \beta \in [60, 90]$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	1.38E-01	8.636	1	2.20E-01	8.046	1	1.48E-01	8.759
$\mu_x^2 + \mu_y^2$	-1.55E-01	8.487	$\mu_x^2 + \mu_y^2$	-3.62E-01	7.487	$\mu_x^2 + \mu_y^2$	-2.05E-01	8.533
$(\bar{v}_{in} - \mu_z)^2$	-4.00E-01	2.315	$(\bar{v}_{in} - \mu_z)^2$	-6.99E-01	2.142	$(\bar{v}_{in} - \mu_z)^2$	-4.48E-01	2.882
μ_z	-1.02E-01	2.070	μ_z	-2.91E-01	1.785	μ_z	-1.93E-01	2.575
$ u_q \mu_z$	-2.28E-02	1.995	$ u_p \mu_y $	-1.84E-02	1.723	$ u_p \mu_z$	4.52E-02	2.418
$ u_p \mu_z$	7.83E-02	1.952	$ u_r \mu_y ^2\mu_z^2$	3.22E+00	1.696	μ_z^3	1.08E+01	2.342
$ u_p \mu_z^2$	5.88E-01	1.940	$ u_q \mu_x $	1.19E-02	1.672	μ_z^4	3.70E+01	2.240
$ u \mu_z^3$	3.48E+00	1.928	$ \bar{r} \mu_y \mu_z^3$	1.34E+03	1.655	μ_z^2	5.82E-01	2.165
$ u_q $	-3.11E-04	1.923	$ u_p \mu_y ^2\mu_z^2$	3.81E+00	1.650	$ u_p \mu_y $	7.06E-03	2.148
R2	0.9503		R2	0.9580		$ u_p \mu_y ^3 \mu_z $	-1.51E+00	2.139
						R2	0.9403	

Table 5 Summary of validation results

	gray-box model			reduced or. hovering* model		
	Output corr.	R2	NRMS (%)	Output corr.	R2	NRMS (%)
F_z	0.9353	0.8636	2.10	0.8831	0.7650	3.03
F_x	0.9945	0.9889	1.54	0.9934	0.9861	1.70
F_y	0.9987	0.9975	1.05	0.9981	0.9961	1.32
M_x	0.7687	0.4847	2.06	0.1944*	-0.0201*	12.63*
M_y	0.8567	0.6883	1.23	0.4141*	-0.0524*	7.53*
M_z	0.8071	0.4873	5.19	0.4152*	0.1417*	13.81*

B. Validation Results

Gray-box models are validated using validation data which are separate from the estimation data but are collected from the same flights. For a better evaluation of these models, the validation outputs are chosen as forces and moments instead of their coefficients.

The force models are compared with the reduced models (25, 32) which have taken into account primary aerodynamic effects. The moment models, however, are compared with the hovering model because no mature reduced model for moment prediction can be found. Metrics of these models are given in Table 5 based on validation results.

Both gray-box and reduced model provide accurate F_x and F_y predictions. From the F_z metrics, it can be concluded that the gray-box model provides better thrust estimations. The RMS of the gray-box force model residuals are reduced by 20-30% on the whole. As for moment predictions, the hovering models (marked by an asterisk) are almost invalid while the gray-box models can provide adequate results. Specifically, the RMS of residuals are reduced by over 80%. To make detailed comparison between above models, several figures are given below.

Fig. 14 gives the validation result of F_z where thrust $T = -F_z$ is plotted for readability. Climbing and descending

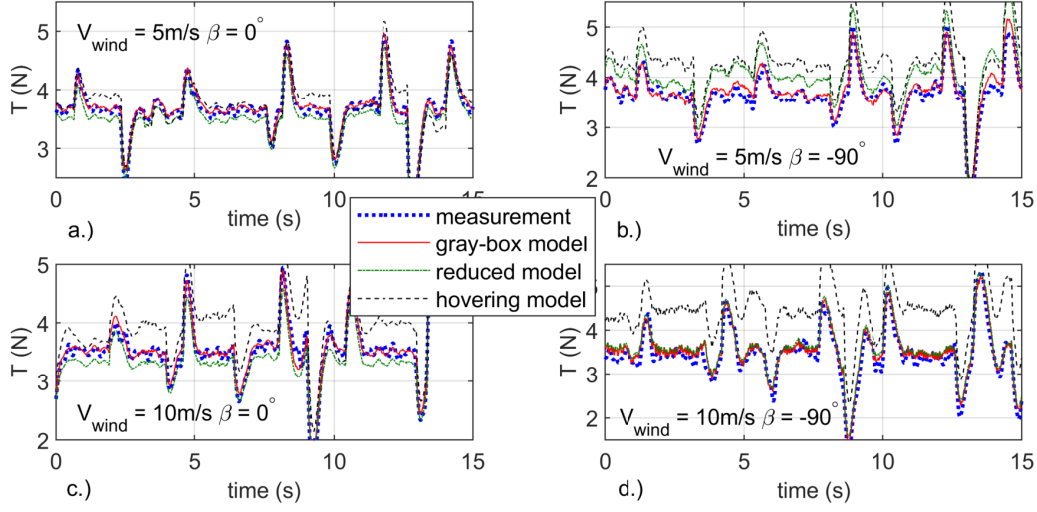


Fig. 14 Validation results of the thrust model. The gray-box model (red solid) is compared with the reduced model (green dash-dot) and the hovering model (black dash). The left two figures illustrate flights when wind speeds are 5 m/s and 10 m/s and the sideslip angle is zero. The right two figures present flights when the sideslip angle is 90° . The gray-box model outperforms the other models in general.

flights are performed at 5 m/s and 10 m/s airspeeds with $\beta = 0$ and $\beta = -90^\circ$ respectively. It's clear that both the gray-box and the reduced model (32) outperform the hovering model, especially in the high-speed and large sideslip flight regime. It is evident that accuracy of the reduced model degrades beyond 5 m/s which could be caused by disturbances from the interactions of the Bebop fuselage with the airflow.

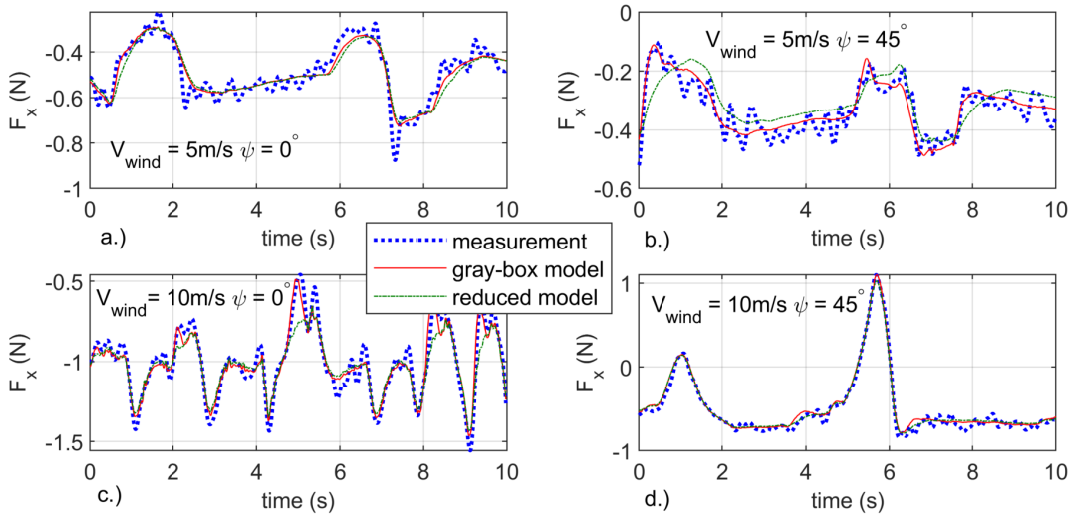


Fig. 15 Validation results of the F_x model. The gray-box model (red solid line) is compared with the reduced model (green dash-dot). Forward and backward flights are performed for validation. The left two figures show flights with zero heading angle, i.e towards the wind tunnel outlet, while the heading angle is 45° in the right figures.

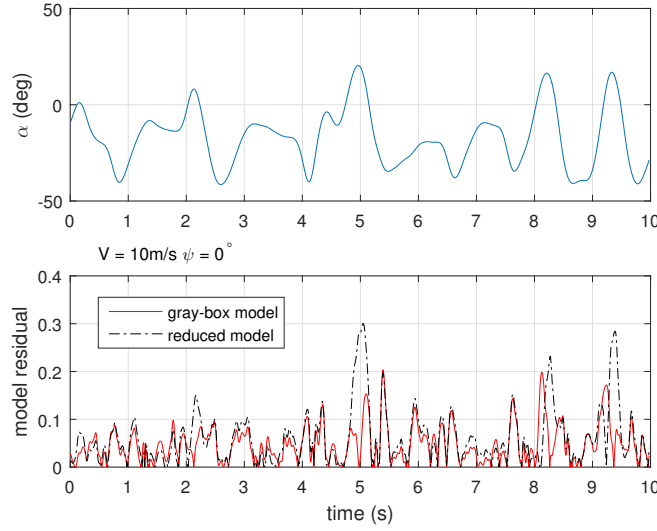


Fig. 16 Time series of angle of attack at $V_{wind} = 10$ m/s and $\psi = 0$. The angle of attack is positive at $t = 5.0$ s and $t = 8.3$ s when errors in the reduced model for F_x appear.

The drag model has been validated by forward and backward flights with different heading angles ψ . Fig. 15 shows the validation results of F_x . The reduced model (25) neglecting interaction effects are compared. Although both models perform well in general, the gray-box model is more accurate in certain parts. For example, at $t = 5.0$ s and $t = 8.3$ s in the Fig. 15b, the gray-box provides accurate prediction while the reduced model produces relatively large errors. The angle of attack and the model residuals of this subplot are given in Fig. 16. It can be seen that the gray-box model outperforms the reduced model when the angle of attack is positive, which means that interactions between the front and aft rotors appear and the reduced model becomes less accurate.

Fig. 17 provides validation results of the F_y model. In flights with 90° heading angles, the quadrotor flew towards the left as what Fig. 3 illustrates. Again, as what is shown in Fig. 18, the gray-box model outperforms the reduced model at points when α is above zero. At $t = 12$ s when angle of attack is positive, a large error appears in predictions made by the reduced model, which does not take into account aerodynamic interaction effects.

Fig. 19 presents the validation result of pitching moment model. The gray-box model outperforms the hovering model as expected. During flights with $\psi = 0$, the prediction of the hovering model is almost always smaller than the measurements since the aft rotors need to rotate much faster than front rotors in the trim condition, which is in line with the observation given in Ref. [22]. This phenomena indicates that during forward flight, a significant pitch up moment appears which is not considered in the hovering model. As for the flights with $\psi = 45^\circ$, although a large aerodynamic coupling exists due to large sideslip angles, the gray-box model can still provide accurate predictions. In addition, validation results of the M_x model are given in Fig. 20 showing the great advantage of the gray-box model.

The model of the yawing moment M_z has been validated by yaw maneuvers and forward-backward maneuvers in the

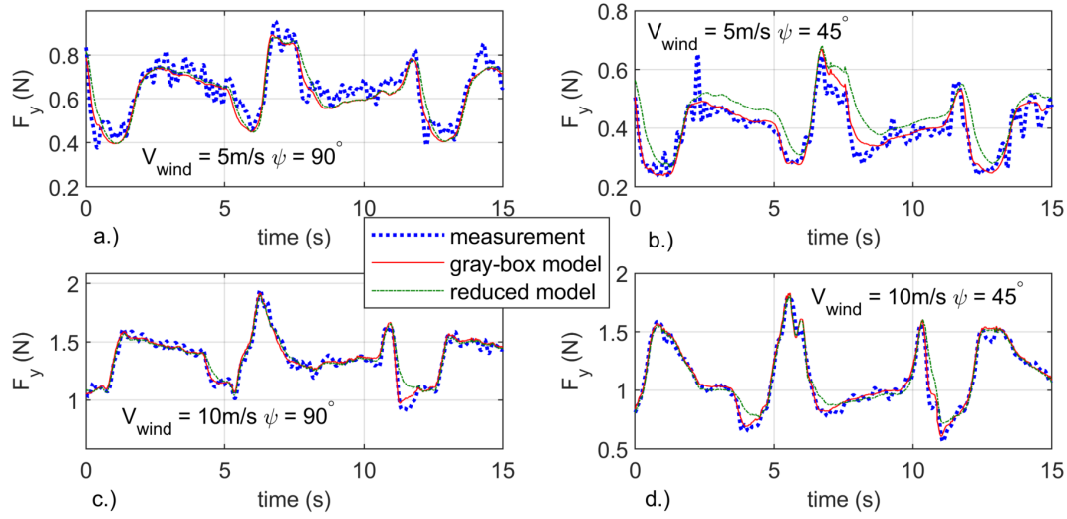


Fig. 17 Validation results of the F_y model. The gray-box model (red solid line) is compared with the reduced model (green dash-dot). Forward and backward flights relative to the wind flow are performed. The left two figures present flights with 90° heading angle, i.e leftward flight against wind flow, while the heading angle is 45° in the right figures.

wind tunnel. The forward-backward maneuvers are carried out with $\psi = 45^\circ$. As can be seen in Fig. 21, predictions from the gray-box model are more accurate than the hovering model. The residual of the hovering model increases as the flight speed grows and a constant bias appears in the Fig. 21d. In this case, the quadrotor flies with $\psi = 45^\circ$, in other words, negative sideslip angle and additional positive yawing moment appear due to aerodynamic effects.

Finally, the gray-box models have been validated near the hovering condition and compared with the hovering models. As Fig. 22 shows, both types of models are accurate. Since aerodynamic resistance is small compared to the presented variables in the hovering condition, models of F_x and F_y are omitted in the plot.

The piecewise polynomial model in this research is discontinuous on the boundary of each section that might be unfavorable for some applications, though the discontinuity can be effectively weakened by increasing the number of model segments. More advanced base functions such as multivariate splines may replace polynomials to guarantee the smoothness of the global model.

VII. Conclusions

Gray-box models of a specific type of quadrotor considering aerodynamic interaction effects have been identified from flight data in a larger flight envelope with respect to the hovering condition. The identification process of this gray-box model consists of the information of phenomenological observation and prior-knowledge about rotorcraft aerodynamics. Therefore, this type of model possesses higher reliability in both high speed and low-speed flight regimes. Although the model structure and parameters are specific to the Bebop platform, the methodology including

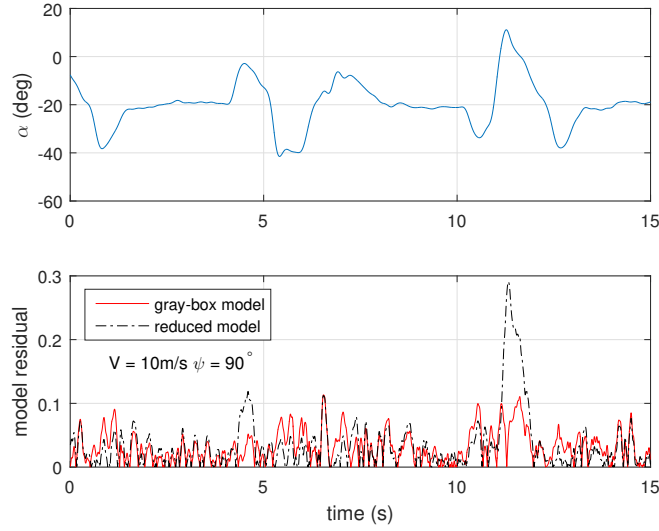


Fig. 18 Time series of angle of attack in $V_{wind} = 10 \text{ m/s}$ and $\psi = 90^\circ$. The angle of attack is positive at $t = 12.0 \text{ s}$ when errors of F_y reduced model appear.

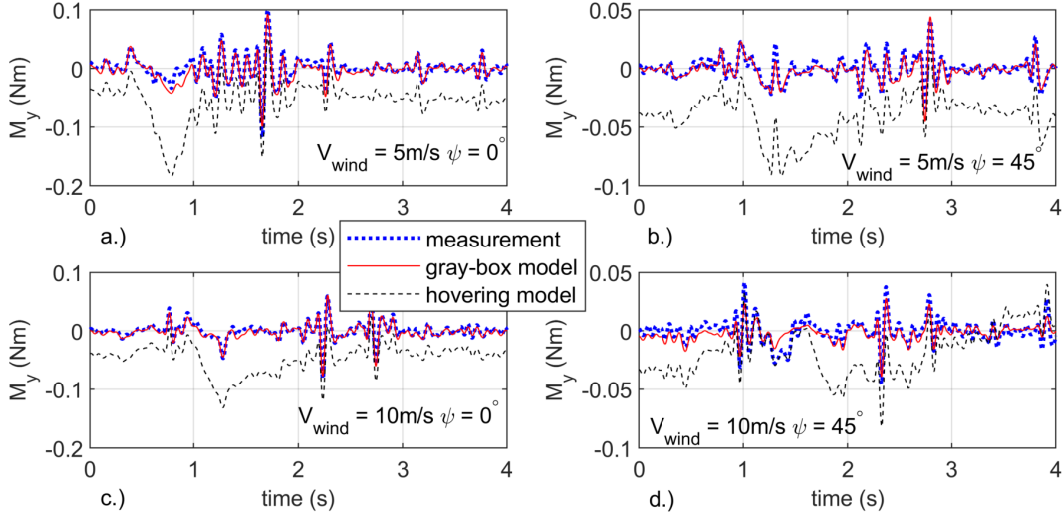


Fig. 19 Validation result of pitching moment M_y model. The gray-box model (red solid) is compared with hovering model (black dash-dot). Forward and backward flights relative to the wind flow are used for validation. The left two figures show flights with zero heading angle, i.e towards the wind tunnel outlet, while heading angle is 45° in the right figures.

nondimensionalization, structure candidates and stepwise regression algorithm, can be generalized to other multi-rotor platforms.

The high-speed flight data have been collected in the wind tunnel. These flight data illustrate significant interaction effects which were rarely considered in the previous literature. The thrust reductions of the aft rotors that obstructed by the front rotors or airframe lead to variations of thrust, moment and even drag force acting on the quadrotor, which may

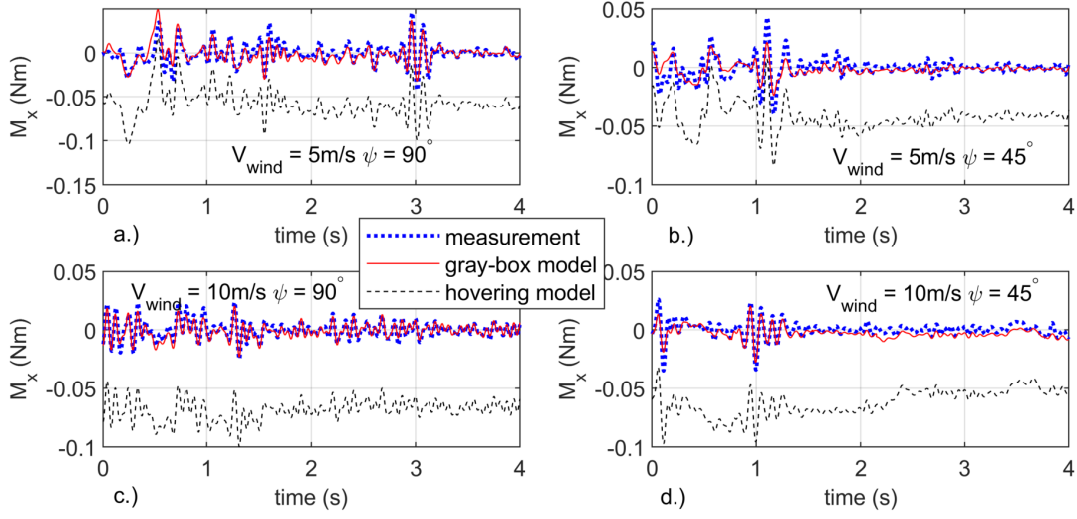


Fig. 20 Validation result of rolling moment M_x model. The gray-box model (red solid) is compared with hovering model (black dash-dot). Forward and backward flights relative to the wind low are performed. The left two figures present flights with 90° heading angle, i.e leftward flight against wind flow, while the heading angle is 45° in the right figures.

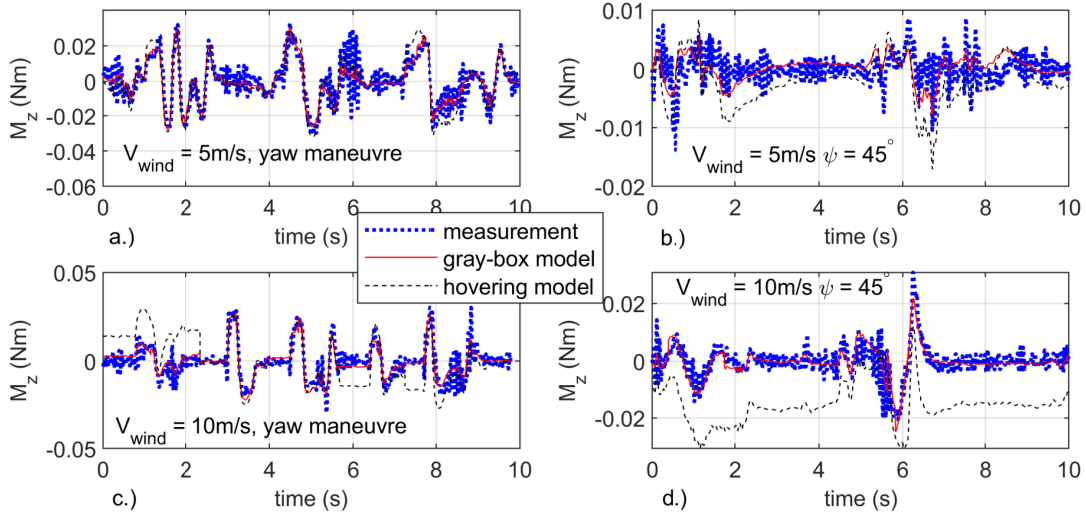


Fig. 21 Validation results of the M_z model. The gray-box model (red solid) is compared with hovering model (black dash-dot). Yaw maneuvers are performed in the left two figures, with different flight speeds. Right two figures present data from forward and backward flights along the wind flow direction with 45° heading angle.

inspire the drone manufacturer to revise their design. For instance, increase the power of aft actuators.

Since the data are obtained from flight tests instead of conventional static wind tunnel tests, and the forces and moments are measured indirectly from on-board and external navigation sensors, this process can also be applied in the open area instead of the wind tunnel. The motion capture system can be replaced by other navigation sensors, such as RTK-GPS, which can produce accurate velocity measurements. Thus the method introduced in this article can be

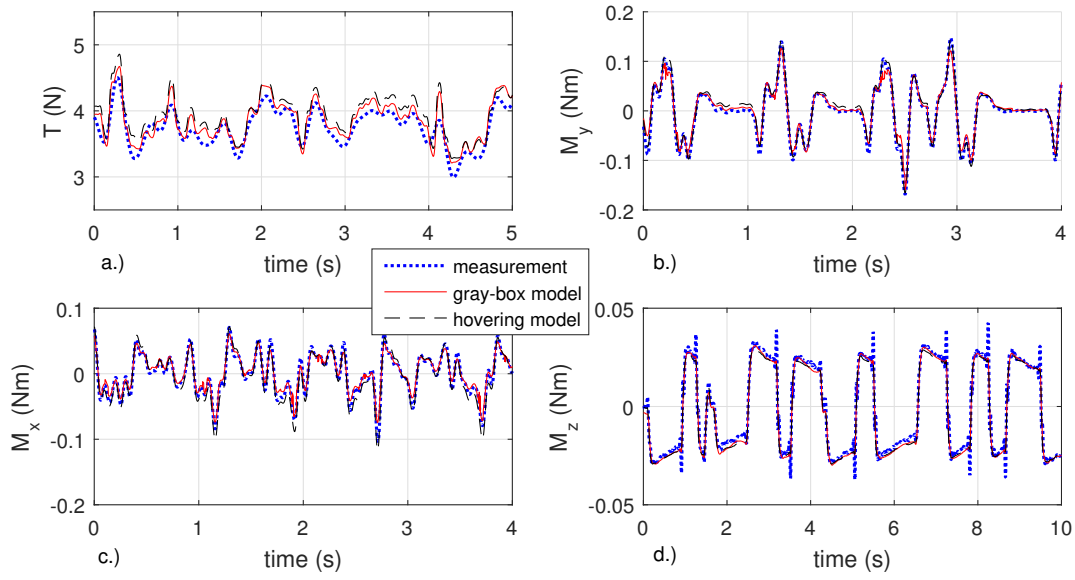


Fig. 22 Validation results during hovering condition. Gray-box model (red solid) is compared with hovering model (green dash-dot). Both model possess enough accuracy in hovering condition.

repeated without wind tunnel equipment to establish accurate models in the interested flight regime.

However, on the other hand, a flight test is only able to explore a limited regime of the flight envelope, which is unfavorable for global model identification. The interaction effects can be only partially revealed by free flight data. Therefore, static wind tunnel tests with force balance are also suggested to carry out for global model identification as well as analyzing interaction effects in detail.

Appendix

A. Stepwise Regression Algorithm

The stepwise regression algorithm is summarized in the Algorithm 1.

B. Estimated Model of C_x , C_y , C_m , C_l and C_n

The estimated aerodynamic coefficients are given in this appendix. Note that the model structure and corresponding parameters are only applicable to Parrot Bebop without bumpers.

Algorithm 1 Forward-backward Stepwise Regression Algorithm

Set initial regressor matrix $\mathbf{A}_0 = [1, 1, \dots, 1]^T \in \mathbb{R}^N$
Set candidate set $\mathcal{S}(y) = \{\xi_0, \xi_1, \dots, \xi_q\}$ containing $q + 1$ candidates
 $k_{max} = 30$; $PSE_{tol} = 10^{-6}$; $F_{out} = 4$; $k = 0$
 $\epsilon_0 = [\mathbf{I} - \mathbf{A}_0(\mathbf{A}_0^T \mathbf{A}_0)^{-1} \mathbf{A}_0^T] \mathbf{z}$
while $k \leq k_{max}$ **do**
 %Forward selection%
 $k = k + 1$
 for $i = 0, 1, \dots, q$ **do**
 $\lambda_i = \xi_i - \mathbf{A}_{k-1}(\mathbf{A}_{k-1}^T \mathbf{A}_{k-1})^{-1} \mathbf{A}_{k-1}^T \mathbf{z}$
 end for
 $j = \operatorname{argmax}_i \operatorname{corr}(\lambda_i, \epsilon_{k-1})^*$ ▷ %corr(x, y) stands for the correlation of x and y%
 $\mathbf{A}_k = [\mathbf{A}_{k-1}, \xi_j]$
 $\hat{\theta} = (\mathbf{A}_k^T \mathbf{A}_k)^{-1} \mathbf{A}_k^T \mathbf{z}$
 $\epsilon_k = \mathbf{z} - \mathbf{A}_k \hat{\theta}_k$
 %Backward elimination%
 Assume there has been p regressors added into $\mathbf{A}_k$
 for $i = 1, 2, \dots, p$ **do**
 Define $\mathbf{A}_{k,i}$ as the reduced regressor matrix of $\mathbf{A}_k$ of which the i th regressor is eliminated
 $\hat{\theta}_{k,i} = (\mathbf{A}_{k,i}^T \mathbf{A}_{k,i})^{-1} \mathbf{A}_{k,i}^T \mathbf{z}$
 $SS_R(\hat{\theta}_k) = \hat{\theta}_k^T \mathbf{A}_k^T \mathbf{z} - N \bar{z}^2$ ▷ %z̄ stands for the mean of z%
 $SS_R(\hat{\theta}_{k,i}) = \hat{\theta}_{k,i}^T \mathbf{A}_{k,i}^T \mathbf{z} - N \bar{z}^2$
 $s^2 = \epsilon_k^T \epsilon_k / (N - p - 1)$
 The F0-ratio of the i th regressor can be calculated by
 $F_{0,i} = [SS_R(\hat{\theta}_k) - SS_R(\hat{\theta}_{k,i})] / s^2$
 end for
 $l = \operatorname{argmin}_i F_{0,i}$
 if $F_{0,l} < F_{out}$ **then** $\mathbf{A}_k = \mathbf{A}_{k,l}$
 $\hat{\theta} = (\mathbf{A}_k^T \mathbf{A}_k)^{-1} \mathbf{A}_k^T \mathbf{z}$
 $\epsilon_k = \mathbf{z} - \mathbf{A}_k \hat{\theta}_k$
 end if
 %Stopping criteria%
 $PSE = \frac{1}{N} \epsilon_k^T \epsilon_k + \frac{p}{N^2} \sum_{i=1}^N [\mathbf{z}(i) - \bar{z}]$
 if $PSE > PSE_{last}$ or $PSE \leq PSE_{tol}$ or $l = j$ **then**
 break
 end if
 $PSE_{last} = PSE$
end while
 $\mathbf{A}_k$ is the final regressors matrix (model structure), $\hat{\theta}_k$ is the estimated parameters and ϵ_k is the model residual.

Table 6 Estimation results of C_x model

$ \beta \in [0, 30]^\circ$			$ \beta \in [30, 60]^\circ$			$ \beta \in [60, 90]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	4.182E-04	18.374	1	4.390E-04	22.466	1	5.007E-04	9.566
μ_x	-3.482E-02	1.327	μ_x	-3.684E-02	2.271	μ_x	-3.873E-02	3.067
$\mu_x \mu_z$	7.717E-02	1.232	$ \mu_y ^3$	7.354E-02	2.213	$ \mu_y ^3$	2.412E-02	2.983
$\mu_x \mu_z^2$	1.057E+00	1.157	$\mu_x \mu_z^2$	3.673E+00	2.190	μ_z	3.956E-03	2.960
$\mu_x \mu_z^3$	3.837E+00	1.119	$\mu_x \mu_z$	2.216E-01	2.111	μ_z^2	3.055E-02	2.915
$\mu_x^2 \mu_z^2$	7.365E-01	1.101	$\mu_x \mu_z^3$	1.497E+01	1.929	$\mu_x \mu_z^2$	2.495E-01	2.904
μ_x^3	-2.883E-02	1.082	μ_x^3	6.234E-02	1.918	R2	0.9079	
μ_z^2	-2.073E-02	1.075	R2	0.9927				
R2	0.9966							

Table 7 Estimation result of C_y model

$ \beta \in [0, 30]^\circ$			$ \beta \in [30, 60]^\circ$			$ \beta \in [60, 90]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	-1.79E-04	6.595	1	-1.98E-04	16.820	1	-1.23E-04	16.563
μ_y	-3.36E-02	2.771	μ_y	-3.54E-02	1.675	μ_y	-3.88E-02	1.283
$\mu_y \mu_z$	1.16E-01	2.718	$ \mu_x \mu_y \mu_z$	2.56E-01	1.201	$\mu_y^3 \mu_z$	4.41E-01	0.979
$\mu_x^2 \mu_z^2$	9.29E-01	2.693	$\mu_y \mu_z^3$	4.64E+00	1.163	$\mu_y \mu_z^3$	1.41E+00	0.945
$ \mu_x \mu_z^2$	-2.10E-01	2.677	$\mu_y \mu_z$	1.61E-01	1.141	$\mu_y \mu_z$	8.39E-02	0.877
$ \mu_x $	5.45E-04	2.664	$\mu_y \mu_z^2$	1.57E+00	1.113	$\mu_y \mu_z^2$	7.41E-01	0.857
R2	0.8368		μ_y^3	-1.05E-01	1.089	μ_y^3	-7.01E-02	0.830
			$ \mu_x $	1.14E-03	1.083	$\mu_y^2 \mu_z^2$	1.12E-01	0.828
			R2	0.9959		R2	0.9975	

Table 8 Estimation result of C_m model, $|\beta| \in [0, 45]^\circ$

$ \beta \in [0, 15]^\circ$			$ \beta \in [15, 30]^\circ$			$ \beta \in [30, 45]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	1.16E-04	6.247	1	5.36E-05	9.779	1	1.73E-05	5.456
u_q	4.44E-03	4.831	u_q	4.70E-03	6.682	u_q	5.01E-03	4.490
$u_q \mu_z$	-5.10E-02	4.589	$u_q \mu_y $	-5.31E-02	4.843	$u_q \mu_y $	-9.56E-02	4.150
μ_x	5.04E-02	4.414	μ_x	4.32E-02	3.473	$u_q \mu_y^2$	8.09E-01	3.168
μ_x^2	1.91E-02	4.019	$\mu_x^3 \mu_z$	5.60E+00	3.093	μ_x	2.84E-02	2.988
μ_z	3.57E-02	3.375	$\mu_x^3 \mu_y^2 \mu_z$	-5.82E+02	2.857	$u_q \mu_z$	-6.23E-02	2.714
$\bar{q}$	-1.35E-01	3.205	$\mu_x \mu_y $	-4.29E-01	2.650	$\mu_x \mu_z^2$	-5.20E+00	2.360
μ_x^3	-1.27E+00	3.072	$u_q \mu_z$	-6.93E-02	2.512	$u_q \mu_y^2 \mu_z$	2.90E+00	2.154
μ_x^4	2.87E+00	2.715	$\bar{q}$	-6.94E-02	2.147	$\mu_x \mu_z^3$	-2.82E+01	1.947
$u_q \mu_x^2 \mu_z$	8.28E-01	2.500	$u_q \mu_z^2$	-3.25E-01	2.088	$\bar{q}$	-8.51E-02	1.768
$u_q \mu_z^3$	9.31E+00	2.410	R2	0.9521		$u_q \mu_x^3 \mu_y^2 \mu_z$	3.53E+02	1.718
$u_q \mu_z^4$	2.25E+01	2.301				$u_q \mu_z^3$	3.43E+00	1.619
$\mu_x^2 \mu_z$	4.11E-01	2.234				$u_q \mu_x^2$	1.46E-01	1.513
$u_q \mu_x \mu_z^3$	-2.31E+01	2.178				$u_q \mu_x^2 \mu_y^2$	-8.85E+00	1.431
$\mu_x^2 \mu_y \mu_z^2$	1.90E+02	2.120				$\mu_y^2 \mu_z^5$	-8.23E+02	1.407
$\mu_x \mu_y^2 \mu_z^2$	-7.62E+02	2.088				R2	0.9393	
R2	0.8872							

Table 9 Estimation result of C_m model, $|\beta| \in [45, 90]^\circ$

$ \beta \in [45, 60]^\circ$			$ \beta \in [60, 75]^\circ$			$ \beta \in [75, 90]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	1.97E-04	3.290	1	-3.82E-05	4.779	1	-1.75E-04	3.590
u_q	4.17E-03	2.837	u_q	4.73E-03	3.767	u_q	3.67E-03	2.635
μ_z	3.00E-03	2.306	μ_x	5.52E-02	3.605	$\mu_y^2 \mu_z$	1.57E-01	2.615
μ_x	3.74E-02	2.128	$\mu_x \mu_z^2$	-7.24E+00	3.400	$u_q \mu_z$	-9.03E-02	2.598
$\mu_x \mu_y \mu_z^2$	7.33E+00	1.995	$u_q \mu_z$	-7.10E-02	3.244	$u_q \mu_z^2$	-9.29E-01	2.562
$u_q \mu_x^2 \mu_y $	2.42E+00	1.900	$\mu_x \mu_y $	-3.57E-01	3.100	$\mu_y^2 \mu_z^3$	-2.56E+01	2.512
$u_q \mu_z$	-6.30E-02	1.845	$u_q \mu_y $	-2.66E-02	2.996	$u_q \mu_y \mu_z^4$	-2.17E+02	2.491
$\mu_x \bar{q}$	-8.64E-01	1.815	$\mu_x \mu_y \mu_z^2$	5.24E+01	2.800	μ_z	-2.44E-03	2.470
$u_q \mu_y \mu_z^2$	-3.67E+00	1.788	$\mu_x \mu_y^2 \mu_z^2$	-1.02E+02	2.641	$ \mu_y \mu_z^5$	2.41E+02	2.448
$\mu_x \mu_z^2$	-5.88E+00	1.706	$\mu_x^3 \mu_y^2 \bar{q}$	-9.88E+03	2.544	$u_q \mu_z^3$	1.49E+01	2.435
$u_q \mu_x^2 \mu_z$	4.35E+00	1.653	$u_q \mu_z^2$	-3.92E-01	2.467	$u_q \mu_z^4$	1.14E+02	2.402
$u_q \mu_x^3 \mu_y^2$	-5.76E+01	1.631	$\mu_x^3 \mu_y \mu_z \bar{q}$	-1.77E+04	2.394	$u_q \mu_y $	-1.06E-02	2.370
$\mu_x \mu_y $	-1.03E-01	1.608	$\bar{q}$	-6.07E-02	2.360	$u_q \mu_y \mu_z^2$	6.79E+00	2.351
$\mu_x \mu_z^3$	-2.71E+01	1.550	$u_q \mu_y^2$	8.98E-02	2.334	$u_q \mu_y \mu_z$	2.45E-01	2.340
$u_q \mu_y $	-2.04E-02	1.517	$u_q \mu_y \mu_z$	3.12E-01	2.315	$\mu_y^2 \mu_z^2$	-2.11E+00	2.307
$u_q \mu_x^3 \mu_y^2 \mu_z$	-5.26E+02	1.476	$u_q \mu_x$	-5.50E-03	2.257	$ \mu_y \mu_z^4$	2.69E+01	2.298
$u_q \mu_x^2 \mu_y \mu_z^2$	8.14E+01	1.438	$\mu_x \mu_y^2$	5.39E-01	2.236	μ_x	1.58E-03	2.292
$u_q \mu_y^2 \mu_z^5$	-1.51E+03	1.425	$\mu_y^2 \mu_z$	8.33E-02	2.185	R2	0.5958	
R2	0.8107		$u_q \mu_y \mu_z^2$	1.41E+00	2.163			
			$\mu_x \mu_z^3$	-5.46E+00	2.156			
			R2	0.7782				

Table 10 Estimation result of C_l model, $|\beta| \in [0, 45]^\circ$

$ \beta \in [0, 15]^\circ$			$ \beta \in [15, 30]^\circ$			$ \beta \in [30, 45]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	2.30E-04	2.940	1	1.28E-04	6.321	1	2.79E-04	6.603
u_p	5.40E-03	1.885	u_p	5.08E-03	3.693	u_p	3.86E-03	6.251
μ_y	-2.84E-02	1.825	μ_y	-1.93E-02	2.695	μ_y	-4.12E-02	5.089
$ \mu_x \bar{p}$	-3.32E-01	1.789	$\mu_x^2\mu_y^3\mu_z$	-9.23E+02	2.186	$\mu_x^2\mu_y$	-2.51E+00	4.567
$\mu_x^2\mu_y$	5.08E-01	1.767	μ_y^2	2.56E-01	2.042	$ \mu_x \mu_y$	5.41E-01	4.407
$u_p \mu_x $	-4.44E-02	1.726	$ \mu_x \mu_y^4$	-1.80E+02	1.795	$\mu_y^3\mu_z$	2.03E+01	4.289
$u_p\mu_z$	-1.26E-02	1.685	$\bar{p}$	-3.72E-02	1.662	$\mu_x^2\mu_y^3$	4.74E+01	4.187
$u_p\mu_x^2$	1.47E-01	1.659	$\mu_z^2\bar{p}$	-8.79E+00	1.637	$\mu_z\bar{p}$	1.33E+00	4.048
$u_p\mu_x^2\mu_z^2$	-2.10E+00	1.629	R2	0.9282		$ \mu_x \mu_y\mu_z^2$	1.54E+01	3.844
$u_p\mu_y$	-4.37E-02	1.607				$u_p\mu_y$	4.73E-03	3.768
$\mu_x^2\mu_y^2$	2.36E+01	1.584				$\mu_x^2\mu_z^5\bar{p}$	2.70E+05	3.741
$u_p\mu_y^2$	3.05E+00	1.550				$\bar{p}$	-4.30E-02	3.720
$u_p\mu_x^2\mu_y^3$	1.75E+03	1.537				$u_p\mu_z^2$	2.64E-02	3.704
$\mu_x^2\mu_y^4$	-8.40E+03	1.519				R2	0.6859	
$\mu_y^2\bar{p}$	-9.54E+01	1.510						
R2	0.7331							

Table 11 Estimation result of C_l model, $|\beta| \in [45, 90]^\circ$

$ \beta \in [45, 60]^\circ$			$ \beta \in [60, 75]^\circ$			$ \beta \in [75, 90]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	3.16E-04	3.626	1	2.69E-04	6.058	1	3.34E-04	3.931
u_p	4.32E-03	3.551	u_p	4.37E-03	5.671	u_p	3.99E-03	3.751
μ_y	-3.45E-02	2.982	μ_y	-2.36E-02	4.600	μ_y	-2.24E-02	3.192
$ \mu_x \mu_y^3\mu_z$	-2.93E+01	2.670	μ_y^3	2.12E-01	3.826	μ_y^3	1.34E-01	2.683
$ \mu_x \mu_y$	4.44E-01	2.592	$\mu_y\mu_z^2$	2.37E+00	3.646	$\mu_y\mu_z^2$	1.35E+00	2.456
$\mu_x^2\mu_y$	-1.54E+00	2.512	$u_p\mu_y^2\mu_z^2$	-3.61E+00	3.473	$\mu_y^2\mu_z\bar{p}$	2.20E+01	2.350
$u_p\mu_z$	1.03E-02	2.475	$u_p\mu_y^2\mu_z$	-6.41E-01	3.369	$\bar{p}$	-5.43E-02	2.187
$ \mu_x \mu_y^2$	-4.09E-02	2.438	$\mu_y^2\bar{p}$	-3.25E+00	3.291	$u_p\mu_z^4$	-4.51E+01	2.159
$\mu_y\mu_z^3$	-7.70E+00	2.414	$u_p\mu_z^5$	-1.66E+02	3.154	$u_p\mu_z^3$	-4.36E+00	2.129
$ \mu_x \mu_y\mu_z^2$	-1.02E+01	2.382	$\mu_y^3\mu_z$	1.31E+00	3.124	μ_z	-2.81E-03	2.087
$\bar{p}$	-5.76E-02	2.352	$u_p\mu_z^4$	-2.39E+01	3.050	$u_p\mu_y^2$	4.01E-02	2.061
μ_y^3	-4.58E-01	2.325	$\mu_y\mu_z^3$	9.49E+00	2.965	$u_p\mu_z^5$	-1.31E+02	2.047
$u_p\mu_z^2$	9.37E-02	2.305	$u_p\mu_z^2$	-2.79E-01	2.922	$\mu_y\mu_z^3$	5.11E+00	2.008
$u_p \mu_x $	-1.96E-02	2.288	$u_p\mu_z^3$	-1.14E+00	2.907	$u_p\mu_y^4$	-3.73E-01	1.994
$u_p\mu_y^2$	6.65E-02	2.269	R2	0.7632		$u_p\mu_z^2$	-1.68E-01	1.975
$\mu_x^2\mu_y^2$	7.15E+00	2.262				μ_z^2	-2.23E-02	1.962
R2	0.6031					R2	0.7476	

Table 12 Estimation result of C_n model, $\mu \leq 0.05$

reg.	$\hat{\theta}$	NRMS (%)
1	2.188E-05	13.930
$s(u_r)u_r^2$	2.032E-04	3.382
r	-3.317E-02	2.689
u_r	3.099E-04	2.492
$\mu_x^2 r$	3.271E+02	2.460
R2	0.9690	

Table 13 Estimation results of C_n model, $\mu > 0.05$

$ \beta \in [0, 30]^\circ$			$ \beta \in [30, 60]^\circ$			$ \beta \in [60, 90]^\circ$		
reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)	reg.	$\hat{\theta}$	NRMS (%)
1	5.470E-05	7.245	1	1.350E-05	7.120	1	2.202E-05	8.291
u_r	6.976E-04	3.446	u_r^3	-4.156E-05	5.883	u_r^3	3.320E-05	6.842
$u_r \mu_x$	-5.354E-04	3.353	$u_r \mu_x$	-3.286E-04	5.471	$\mu_x \mu_y$	-4.661E-02	6.280
$\mu_y \mu_z$	4.554E-02	3.276	$\mu_x \mu_y^3$	-2.113E-01	5.368	$s(u_r)u_r^2$	-7.975E-05	6.170
$s(r)\bar{r}^2$	-1.781E+00	3.244	μ_y^3	-2.595E-02	5.100	μ_y	-1.640E-03	5.702
R2	0.8060		$s(u_r)u_r^2$	3.005E-04	4.485	u_r	4.716E-04	5.445
			$\mu_x \mu_x$	1.349E-02	4.413	$s(\bar{r})\bar{r}^2$	-2.994E+00	5.366
			$s(\mu_y)\mu_y^2\mu_z$	2.348E-01	4.335	$\mu_x \mu_y \mu_z$	-1.210E-01	5.352
			$u_r \mu_z$	-4.223E-03	4.245	R2	0.5859	
			$s(\bar{r})\bar{r}^2 u_r$	5.652E+01	4.188			
			$s(u_r)u_r \mu_z$	1.287E-03	4.156			
			$\mu_y \mu_z^2$	1.188E-01	4.141			
			R2	0.6652				

References

- [1] Mellinger, D., Michael, N., and Kumar, V., “Trajectory Generation and Control For Precise Aggressive Maneuvers with Quadrotors,” *Springer Tracts in Advanced Robotics*, Vol. 79, 2014, pp. 361–373. doi:10.1007/978-3-642-28572-1_25.
- [2] Hehn, M., and Dandrea, R., “Real-Time Trajectory Generation for Quadcopters,” *IEEE Transactions on Robotics*, Vol. 31, No. 4, 2015, pp. 877–892. doi:10.1109/TRO.2015.2432611.
- [3] Smeur, E. J., De Croon, G. C., and Chu, Q., “Gust Disturbance Alleviation with Incremental Nonlinear Dynamic Inversion,” *IEEE International Conference on Intelligent Robots and Systems*, Vol. 2016-Novem, 2016, pp. 5626–5631. doi:10.1109/IROS.2016.7759827.
- [4] Huang, H., Hoffmann, G. M., Waslander, S. L., and Tomlin, C. J., “Aerodynamics and control of autonomous quadrotor helicopters in aggressive maneuvering,” *Proceedings - IEEE International Conference on Robotics and Automation*, 2009, pp. 3277–3282. doi:10.1109/ROBOT.2009.5152561.
- [5] Alexis, K., Nikolakopoulos, G., and Tzes, A., “Switching Model Predictive Attitude Control for a Quadrotor Helicopter Subject to Atmospheric Disturbances,” *Control Engineering Practice*, Vol. 19, No. 10, 2011, pp. 1195–1207. doi:10.1016/j.conengprac.2011.06.010.
- [6] Leishman, R. C., MacDonald, J. C., Beard, R. W., and McLain, T. W., “Quadrotors and Accelerometers: State Estimation with an Improved Dynamic Model,” *IEEE Control Systems*, Vol. 34, No. 1, 2014, pp. 28–41. doi:10.1109/MCS.2013.2287362.
- [7] Carroll, T., George, I.-R. E., and Bramesfeld, G., “Design Optimization of Small Rotors in Quad-Rotor Configuration,” *54th AIAA Aerospace Sciences Meeting*, 2016. doi:10.2514/6.2016-1788.
- [8] Foster, J. V., and Hartman, D., “High-Fidelity Multi-Rotor Unmanned Aircraft System (UAS) Simulation Development for Trajectory Prediction Under Off-Nominal Flight Dynamics,” *17th AIAA Aviation Technology, Integration, and Operations Conference*, 2017. doi:10.2514/6.2017-3271.
- [9] Zhang, Y., de Visser, C. C., and Chu, Q. P., “Aircraft Damage Identification and Classification for Database-Driven Online Flight-Envelope Prediction,” *Journal of Guidance, Control, and Dynamics*, Vol. 41, No. 2, 2017, pp. 449–460. doi:10.2514/1.G002866.
- [10] Hoffmann, G. M., Huang, H., Waslander, S. L., and Tomlin, C. J., “Quadrotor Helicopter Flight Dynamics and Control : Theory and Experiment,” *American Institute of Aeronautics and Astronautics*, Vol. 4, No. August, 2007, pp. 1–20. doi:10.2514/6.2007-6461.
- [11] Hoffmann, G. M., Huang, H., Waslander, S. L., and Tomlin, C. J., “Precision Flight Control for a Multi-Vehicle Quadrotor Helicopter Testbed,” *Control Engineering Practice*, Vol. 19, No. 9, 2011, pp. 1023–1036. doi:10.1016/j.conengprac.2011.04.005.
- [12] Martin, P., and Salaün, E., “The True Role of Accelerometer Feedback in Quadrotor Control,” *Proceedings - IEEE International Conference on Robotics and Automation*, 2010, pp. 1623–1629. doi:10.1109/ROBOT.2010.5509980.

- [13] Kaya, D., and Kutay, A. T., "Aerodynamic Modeling and Parameter Estimation of a Quadrotor Helicopter," *AIAA Atmospheric Flight Mechanics Conference*, 2014. doi:10.2514/6.2014-2558.
- [14] Orsag, M., and Bog, S., "Influence of Forward and Descent Flight on Quadrotor Dynamics," *Recent Advances in Aircraft Technology*, InTech, 2012, pp. 141–156. doi:10.5772/37438.
- [15] Tang, Y. R., and Li, Y., "Dynamic Modeling for High-Performance Controller Design of a UAV Quadrotor," *2015 IEEE International Conference on Information and Automation*, 2015, pp. 3112–3117. doi:10.1109/ICInfA.2015.7279823.
- [16] Schulz, M., Augugliaro, F., Ritz, R., and D'Andrea, R., "High-Speed, Steady Flight with a Quadcopter in a Confined Environment Using a Tether," *IEEE International Conference on Intelligent Robots and Systems*, Vol. 2015-Decem, 2015, pp. 1279–1284. doi:10.1109/IROS.2015.7353533.
- [17] Gill, R., and Andrea, R. D., "Propeller Thrust and Drag in Forward Flight," *IEEE Conference on Control Technology and Applications (CCTA)*, Mauna Lani, HI, 2017, pp. 73–79. doi:10.1109/CCTA.2017.8062443.
- [18] Khan, W., and Nahon, M., "Toward an Accurate Physics-based UAV Thruster Model," *IEEE/ASME Transactions on Mechatronics*, Vol. 18, No. 4, 2013, pp. 1269–1279. doi:10.1109/TMECH.2013.2264105.
- [19] Mahony, R., Kumar, V., and Corke, P., "Multirotor Aerial Vehicles: Modeling, Estimation, and Control of Quadrotor," *IEEE Robotics & Automation Magazine*, Vol. 19, No. 3, 2012, pp. 20–32. doi:10.1109/MRA.2012.2206474.
- [20] Bristeau, P.-j., Martin, P., Salaun, E., and Petit, N., "The role of propeller aerodynamics in the model of a quadrotor UAV," *2009 European Control Conference (ECC)*, IEEE, 2009, pp. 683–688. doi:10.23919/ECC.2009.7074482.
- [21] Pounds, P., Mahony, R., and Corke, P., "Modelling and Control of a Large Quadrotor Robot," *Control Engineering Practice*, Vol. 18, No. 7, 2010, pp. 691–699. doi:10.1016/j.conengprac.2010.02.008.
- [22] Russell, C., Jung, J., Willink, G., and Glasner, B., "Wind Tunnel and Hover Performance Test Results for Multicopter UAS Vehicles," *American Helicopter Society 72nd Annual Forum.*, West Palm Beach, FL, 2016.
- [23] Luo, J., Zhu, L., and Yan, G., "Novel Quadrotor Forward-Flight Model Based on Wake Interference," *AIAA Journal*, Vol. 53, No. 12, 2015, pp. 3522–3533. doi:10.2514/1.J053011.
- [24] Morelli, E. A., "Global Nonlinear Aerodynamic Modeling Using Multivariate Orthogonal Functions," *Journal of Aircraft*, Vol. 32, No. 2, 1995, pp. 270–277. doi:10.2514/3.46712.
- [25] Klein, V., and Morelli, E. A., *Aircraft System Identification: Theory and Practice*, AIAA, Blacksburg, VA, 2006. doi: 10.2514/4.861505.
- [26] Lombaerts, T., "Fault Tolerant Flight Control: A Physical Model Approach," Ph.D. thesis, Delft Univ. of Technology, Delft, The Netherlands, 2010.

- [27] Armanini, S. F., Karásek, M., de Croon, G. C. H. E., and de Visser, C. C., “Onboard/Offboard Sensor Fusion for High-Fidelity Flapping-Wing Robot Flight Data,” *Journal of Guidance, Control, and Dynamics*, Vol. 40, No. 8, 2017, pp. 2121–2132. doi:10.2514/1.G002527.
- [28] Smeur, E. J. J., Chu, Q., and Guido C. H. E. de Croon, “Adaptive Incremental Nonlinear Dynamic Inversion for Attitude Control of Micro Aerial Vehicles,” *Journal of Guidance, Control, and Dynamics*, Vol. 39, No. 3, 2016, pp. 450–461. doi:10.2514/1.G001490.
- [29] “Paparazzi open-source unmanned (air) vehicle project,” , Apr 2018. URL <https://github.com/paparazzi/paparazzi>.
- [30] Mendes, A., van Kampen, E., Remes, B., and Chu, Q. P., “Determining moments of inertia of small UAVs: A comparative analysis of an experimental method versus theoretical approaches,” *AIAA Guidance, Navigation, and Control Conference*, American Institute of Aeronautics and Astronautics, Reston, Virigina, 2012. doi:10.2514/6.2012-4463.
- [31] Powers, C., Mellinger, D., Kushleyev, A., Kothmann, B., Kumar, V., *Influence of Aerodynamics and Proximity Effects in Quadrotor Flight*, Springer Tracts in Advanced Robotics, Springer International Publishing, Heidelberg, 2013. doi: 10.1007/978-3-319-00065-7.